C_4-Free Subgraphs of the Hypercubes Q_6, Q_7, and Q_8: Odd Squares, Fully Frustrated Models, and Computational Structure

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Independent

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This is a revision of a manuscript first circulated as arXiv:2603.29127 (v1–v4, March–May 2026); v1 appeared as "New Lower Bounds for C_4-Free Subgraphs of the Hypercubes Q_7 and Q_8", and by v4 the title had become "New Lower Bounds for C_4-Free Subgraphs of the Hypercubes Q_6, Q_7, and Q_8: Constructions, Structure, and Computational Method." That earlier version announced $\mathrm{ex}(Q_8, C_4) \geq 680$ and did not include the odd-square material of Section 6; see the Acknowledgements for the circumstances of the revision. Three further claims of v1–v4 are withdrawn or corrected here, and are recorded as such rather than quietly dropped. (i) Its Section 7 was titled "Computational *Proof* that $\mathrm{ex}(Q_6, C_4) = 132$" and asserted that standard MIP solvers verify the instance *quickly*; we were unable to close the optimality gap (Section 9), so that section is retitled and the upper bound $\mathrm{ex}(Q_6, C_4) \leq 132$ again rests solely on Harborth–Nienborg [11]. (ii) Its abstract gave the pairwise Hamming range of the Q_7 catalogue as 36–260; that came from a 5 000-pair sample, disclosed only in the body. Exhaustive computation over all 197 319 045 pairs gives 6–274 (Section 7.3), so both endpoints were substantially wrong and the inference that the solutions lie far apart does not survive. (iii) Its degree sequence for the 132-edge Q_6 construction, $\{4^{32}, 5^{32}\}$, is arithmetically impossible — the degree sum is 288, i.e. 144 edges — and the correct sequence is $\{4^{56}, 5^{8}\}$ (Section 8.5); the "degree support $\{n-3, n-2, n-1\}$" regularity suggested there rested on that error and is withdrawn. The extrapolated Brass–Harborth–Nienborg entries for $n = 4, 5, 6$ in v1–v4's value table are likewise replaced by explicit out-of-domain markers (Table 1).

Abstract

We study C_4-free subgraphs of the hypercubes Q_6, Q_7 and Q_8. Writing $g(n)$ for the maximum size of an *odd-square* edge set — one meeting every 4-cycle (*square*) of Q_n in exactly one or three edges — we prove $g(6) = 132$, $g(7) = 304$ and $g(8) = 682$, give matching explicit machine-verifiable edge-list certificates, and hence obtain $\mathrm{ex}(Q_7, C_4) \geq 304$ and $\mathrm{ex}(Q_8, C_4) \geq 682$. Under the standard correspondence with fully frustrated signed hypercubes these values are already implicit in the statistical-physics literature: the field identity, the integer optimisation it feeds, and its optimal values go back to Derrida, Pomeau, Toulouse and Vannimenus (1979), who also construct ground states for $D \leq 7$; the $D = 8$ attainment was reported, without any released configuration, by Marinari, Parisi and Ritort (1995); and Laplante, Naserasr, Sunny and Wang treat the graph-theoretic formulation directly. Our contribution is accordingly narrow and explicit: a short self-contained proof of the optima via a mod-8 refinement of the classical two-point inequality; the certificates themselves, with dependency-free verifiers; and an exhaustive decision of the realisability of the optimal $n = 6$ field distributions — settling the question DPTV left open in 1979 about their "other solutions" — where an independent integer programme shows the optimal distributions number exactly three, and the decision shows exactly one of them is realisable (the other two are not, upgrading MPR's tentative single-case report to a proof covering both). Full attribution, with primary-source quotations and our translation conventions, is spelled out in Sections 2 and 6.

Independently of the odd-square results, we classify the 19 866 previously released 304-edge C_4-free subgraphs of Q_7 (exactly 389 of them odd-square) into 20 dimension-profile types, establish their common structural core (degree sequence $\{4^{32}, 5^{96}\}$, spectral radius in $[4.78543, 4.79129]$, local maximality — properties of the released catalogue, not claims about all 304-edge solutions), and decompose them into 180 $\mathrm{Aut}(Q_7)$-orbits containing 34 227 200 labelled solutions, of which the released files are 0.058%; whether these are *all* orbits of 304-edge solutions is open. This revision adds an orbit-witness certificate whose standard-library checker verifies the orbit *assignment* in seconds (hence "at most 180 orbits"); its `--canonical` mode certifies "exactly 180" independently of the census scan. For Q_6, the unrestricted value $\mathrm{ex}(Q_6, C_4) = 132$ rests on Harborth–Nienborg's combinatorial upper bound, which we did not independently verify; the odd-square $g(6) = 132$ is self-contained here. Every numerical claim in the certificates and in our own computational analyses is re-checkable from the released code and data (the one imported exact result — the unrestricted $n = 6$ upper bound — is marked as external throughout); the provenance record separates such claims from purely historical statements, and it, the repository, and its SHA-256 manifests are collected in Section 2. Edge lists and code:
`https://github.com/minamominamoto/c4free-hypercube`.

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Contents

1 Introduction

The n-dimensional hypercube Q_n has vertex set $\{0,1\}^n$, two vertices adjacent iff they differ in exactly one coordinate; it has 2^n vertices and $n \cdot 2^{n-1}$ edges. The problem of determining $\mathrm{ex}(Q_n, C_4) = \max\{|E(G)| : G \subseteq Q_n,\ G \text{ is } C_4\text{-free}\}$ was raised by Erdős [9] and remains open in general.

The best known asymptotic bounds are

$$\tfrac{1}{2}(n+\sqrt{n})\cdot 2^{n-1} \;\le\; \mathrm{ex}(Q_n, C_4) \quad (n = 4^r), \qquad \mathrm{ex}(Q_n, C_4) \;\le\; (0.60318+o(1))\cdot n\cdot 2^{n-1}, \quad (1)$$

where the lower bound is due to Brass, Harborth, and Nienborg [3] and holds, in this form, only when n is a power of 4; the side condition is attached to the displayed inequality rather than left to the surrounding text, since the two halves of (1) have different scopes. The upper bound is a statement about the edge Turán *density* $\pi_e = \limsup_n \mathrm{ex}(Q_n, C_4)/(n2^{n-1})$, not a pointwise inequality valid at every finite n; the constant 0.60318 is due to Baber [1], who used the semidefinite flag algebra method to improve the earlier bound of Thomason and Wagner [16]; the intermediate value 0.6068 is due to Balogh, Hu, Lidický, and Liu [2]. For general $n \ge 9$, the weaker bound $\frac{1}{2}(n + 0.9\sqrt{n}) \cdot 2^{n-1}$ applies [3]. A wording caution is useful for cross-checking the literature: LNSW26's discussion of this BHN expression contains a sentence calling it an "at most" bound (the sentence persists in the published version's Introduction, which we checked directly), whereas BHN95's own statement gives both expressions as lower bounds on the maximum edge count, which settles the direction at the source. We flag this explicitly as a direction slip in that secondary description rather than an alternative theorem: an "at most" reading is internally untenable at $n = 16$, where the tight $n = 4^r$ form gives $(16+4)\cdot 2^{14} = 327\,680$, exceeding $(16+0.9\cdot 4)\cdot 2^{14} = 321\,126.4$. We follow BHN95. Exact values have been determined for small n; Harborth and Nienborg [11] proved $\mathrm{ex}(Q_6, C_4) = 132$.

Our main results, stated in certificate form – as the attribution immediately below and Section 6 spell out, the new content is the explicit machine-verifiable certificates and the graph-theoretic translation, not the bound values themselves:

Theorem 1 (Certified lower bounds). *(i)* $\mathrm{ex}(Q_7, C_4) \geq 304$.

(ii) $\mathrm{ex}(Q_8, C_4) \geq 682$.

Part (i) is not new: it follows from the 1979 construction of Derrida, Pomeau, Toulouse, and Vannimenus [5] (Section 6), and we present it here as an independently reproduced and machine-verified result rather than as a novel bound. Part (ii) improves our originally announced bound $\mathrm{ex}(Q_8, C_4) \geq 680$, but here too the bare existence claim is not new in substance: Marinari, Parisi, and Ritort [15] report, in a single unelaborated sentence and with no accompanying spin configuration or machine-checkable certificate, having exhibited a $D = 8$ ground state attaining Derrida et al.'s improved energy lower bound; they describe the method (simulated annealing with a cooling procedure) and state the corresponding bound value as -361 on the D=9 comparison scale they use for cross-dimension bound values — a normalised per-spin energy multiplied by their D=9 factor of 768, not a count of couplings or any other absolute integer of the D=8 model itself; Section 6 gives the full conversion (in their text the attainment statement and this integer occur in two separate sentences, combined here), but provide no graph-theoretic translation and no reproducible computational artifact. Combined with the signed-graph/odd-square correspondence and edge-count–field-sum translation we give in Section 6, this would imply the existence of a 682-edge odd-square subgraph of Q_8 – but as an existence claim with no public certificate, it is not independently verifiable, and we do not treat it as settling the question on its own. What Part (ii) contributes beyond that prior existence claim is: an explicit graph-theoretic translation of MPR's result into edge-count terms (Section 6.2); a publicly released, explicit, machine-verifiable 682-edge certificate, which MPR's paper does not supply (they report the result in energy units only, without an explicit spin configuration); and the structural analysis of that certificate (Proposition 15). The specific certificate we release attains the same bound value MPR report; its own provenance is recorded as a dated private-correspondence account rather than an independently reproducible computational log (see Section 6.3); we do not claim it reproduces any particular configuration MPR may have used internally. That asymmetry — the headline certificate being the one artefact whose *derivation* could not be re-executed — is closed in this revision by `q8_A_recover.py`, a fixed-seed search that finds, with no input from the released witness, a spin configuration realising the same optimal field distribution (seed 90008, hit at iteration 207 579 *at the script's default iteration budget*: the budget parameter enters the annealing temperature schedule, so this exact hit – and the byte-identical output, which `reproduce_core.py` now asserts by SHA-256 – is conditional on the default 250 000-iteration setting, a different budget yielding a different, equally valid witness, and is scoped to the reference platform class: the acceptance step evaluates a floating-point exponential, so the byte-identity is an observed stability of CPython on IEEE-754 hardware with a standard `libm` (reproduced in our CPython 3.12 environment and, via an RNG-faithful independent re-implementation, by a reviewer), not a universal cross-platform guarantee — and it concerns only the byte-level identity of the *regenerated* file: the existence of a 682-edge odd-square subgraph is carried platform-independently by the released static certificate and its dependency-free verification; 682 edges, $h|_U = \{0{:}1, 2{:}84, 4{:}43\}$, square histogram $\{1{:}301, 3{:}1491\}$, released as `q8_A_witness.json`). It is a different edge set from the released witness, as expected; what is now reproducible is the existence, not the particular artefact. Both bounds are realised by explicit constructions (supplementary files `q7_edges_304.jsonl.part1--3` and `q8_odd_square_682.json`; the originally announced 680-edge Q_8 witness, `q8_edges_680.jsonl`, is retained as a simulated-annealing data point, see Section 5).

Within the *odd-square* subclass — edge sets meeting every square in an odd number of edges — the extremal value $g(n)$ is determined exactly for $n = 6, 7, 8$:

Theorem 2 (Exact odd-square values)**.** *$g(6) = 132$, $g(7) = 304$, and $g(8) = 682$. The numerical values for $n = 6, 7$ have historical priority in Derrida–Pomeau–Toulouse–Vannimenus [5], and MPR95 [15] reported attainment of the corresponding $n = 8$ ground-state bound. The assertion here is a self-contained graph-theoretic re-proof/certification of those values, not a claim that the three numbers themselves are new.*

To keep the priority record in one place rather than scattered across the Abstract, this introduction, Section 6, and Open Problem 8, the attribution of the three values is:

n	value $g(n)$: priority	contribution here
6	attained by DPTV79 (D=6 ground state); value attributed to LNSW26 by [13]	certificate; short proof
7	DPTV79 (explicit D=7 construction)	certificates; short proof
8	MPR95 (attainment reported in energy units, no data)	682-edge translation; certificate; short proof

Here "short proof" means Lemma 12's self-contained derivation of the optima DPTV tabulate; in no row are the numerical values themselves claimed as new. One caveat travels with the $n = 6$ row: DPTV79's Table I could not be text-extracted from the scans we obtained, so the identification of its "other solutions" with our distributions A and B rests on the completeness of our own enumeration together with the surrounding prose, not on a direct transcription (disclosed in Section 6); the exhaustive realisability decision itself is independent of that reading.

All three values are self-contained within the released materials: each upper bound follows from Lemma 9 and Lemma 12 (Section 6.2), and each lower bound is witnessed by an explicit, machine-verified odd-square edge list released with this paper (Section 6.3). The citations to [5, 15] record historical priority for the values, not a dependency of the proofs. One point of possible confusion: the released 132-edge Q_6 construction `q6_edges_132.jsonl` is a genuine, independently machine-verified C_4-free witness for $\mathrm{ex}(Q_6, C_4) \geq 132$, but it is *not* odd-square; the odd-square witness for $g(6) \geq 132$ is the separate file `q6_odd_square_132.json`.

Theorem 2 is proved in Section 6. Combined with $g(n) \leq \mathrm{ex}(Q_n, C_4)$, it in fact implies Theorem 1's lower bounds $\mathrm{ex}(Q_7, C_4) \geq 304$ and $\mathrm{ex}(Q_8, C_4) \geq 682$ directly (redundantly with the explicit constructions of Sections 4–5). What it does *not* resolve is whether these bounds are exact, i.e. whether $\mathrm{ex}(Q_7, C_4) = 304$ and $\mathrm{ex}(Q_8, C_4) = 682$: a general C_4-free subgraph need not be odd-square, so $g(n) = \mathrm{ex}(Q_n, C_4)$ does not follow automatically, and of the 19 866 published 304-edge Q_7 solutions, only 389 are odd-square (Section 7).

Four caveats attach to Table 1. First, Eq. (1)'s tight form applies only for $n = 4^r$ and its weaker form only for $n \geq 9$ [3], so the $n = 5, 6$ entries are not covered by either stated

Table 1: Known values and lower bounds of $\mathrm{ex}(Q_n, C_4)$. The BHN column gives Brass–Harborth–Nienborg estimates from Eq. (1); the domain caveats attached to that column, the status of the $n = 4$ entry, and the meaning of the Source column are spelled out in the text immediately below the table.

n	$\lvert E(Q_n)\rvert$	$\mathrm{ex}(Q_n, C_4)$	BHN (Eq. 1)	Source
4	32	24	24 (tight, $n = 4^1$)	elementary (proof below); construction via [3]
5	80	56	n/a (outside domain)	Emamy-K et al. [6]
6	192	$132^{\ddagger}$	n/a (outside domain)	Harborth–Nienborg [11]
7	448	$\geq$ **304**	outside domain†	value attained by [5]; witness ours
8	1024	$\geq$ **682**	outside domain†	bound-attainment reported by [15] (energy units); 682-edge translation and witness ours

form (marked n/a) and the $n = 7, 8$ entries lie *beyond* the stated domain of both forms (marked †): the out-of-domain extrapolations of the $n \geq 9$ form evaluate there to ≈ 300.2 and ≈ 674.9 respectively, figures recorded here only, for illustrative comparison, never as proven bounds; no claim in this paper depends on them. Second, the $n = 4$ value admits a two-line proof, recorded here because we could not locate an explicit page-level statement of it in the literature (not in [4], and neither OEIS A245762 nor the Emléktábla collection attributes the $n = 4$ entry individually): Q_4 has $\binom{4}{2} \cdot 2^2 = 24$ squares and each edge lies in exactly $n - 1 = 3$ of them, so a C_4-free subgraph must omit at least one edge of every square and hence omit at least $\lceil 24/3 \rceil = 8$ of the 32 edges, giving $\mathrm{ex}(Q_4, C_4) \leq 24$; the matching construction is the $n = 4^r$ case ($r = 1$) of Brass–Harborth–Nienborg [3], whose lower bound $\frac{1}{2}(n + \sqrt{n})2^{n-1}$ evaluates to exactly 24 there. Third, in the Source column, "value; witness ours" means the numerical value has historical priority in the cited work, which published no machine-readable edge list or spin configuration for it; the explicit certificate released here is the present author's own and is not claimed to reproduce any configuration of the cited work. Fourth (‡), the $n = 6$ upper bound $\mathrm{ex}(Q_6, C_4) \leq 132$ is used here on the authority of [11] and has not been independently re-verified in this work (our own ILP attempt did not close the optimality gap, Section 9.2); the odd-square value $g(6) = 132$ is self-contained and unaffected.

The paper is organised as follows. Section 2 collects scope, attribution, the verification map, and the provenance record in one place (material that earlier revisions carried in the abstract). Section 3 describes the C_4 enumeration and verification procedure used throughout. Sections 4–5 present the originally announced Q_7 and Q_8 simulated-annealing constructions. Section 6 introduces the odd-square subclass, proves Theorem 2, and presents the reconstructed 682-edge Q_8 witness and its verification. Section 7 classifies the 19 866 Q_7 solutions found by their dimension profiles, yielding exactly 20 types, and reports the odd-square audit (389 of 19 866). Section 8 describes the two-phase simulated annealing algorithm. Section 9 reproduces $\mathrm{ex}(Q_6, C_4) = 132$ computationally on the lower-bound side, with an ILP formulation for the upper-bound side. Section 10 lists open problems.

2 Reader's Guide: Scope, Attribution, and Provenance

This section collects in one place the scope limitations, the attribution record with primary-source detail, the map of what is certified and how to check it, and the provenance of the released constructions. Earlier revisions carried this material in the abstract; the abstract now summarises, and this section is the authoritative expanded statement. Nothing here is new relative to the previous revision except where explicitly marked; the text is relocated, not retracted. One reading convention, stated once here instead of piecemeal: several quantities reported in this paper (for example the square split in Proposition 13(ii), the $\mathrm{Tr}(A^4)$ value in Proposition 19(iii), and the $h \geq 0$ restriction of Lemma 11) are arithmetically forced by the edge count, the degree sequence, or the odd-square condition rather than independent findings; where this happens the text marks them as consistency checks, and they should be read as such, not as separate results. A second convention of the same kind: every *certified* mathematical claim in this paper depends only on artefacts bundled with the release; the items individually marked as reported-but-not-auditable (S. Lai's independent cross-check, the recovered 72-hour solver log, the SAT/MIP trial statistics, and MPR95's energy inputs) support the provenance and attribution narrative only, never a theorem.

After this manuscript was first circulated, a reader (S. Lai) identified an overlooked connection to a line of statistical-physics literature on fully frustrated hypercubes: under the standard correspondence between edge sets meeting every 4-cycle (*square*) an odd number of times and fully frustrated signed hypercubes, the 1979 construction of Derrida, Pomeau, Toulouse, and Vannimenus [5] already yields a $D \leq 7$ odd-square construction attaining 304 edges at $D = 7$ (we did not reconstruct their specific H_7 edge set), and the D=8 ground state reported by Marinari, Parisi, and Ritort [15] yields a 682-edge C_4-free subgraph of Q_8 that improves our originally announced 680-edge construction. For Q_7, this value (304) is also attained by 389 of our own previously found 304-edge solutions, which we confirm independently satisfy the odd-square condition (none is claimed to be DPTV's specific H_7 construction; we only confirm the attained value, not edge-set identity with theirs). For Q_8, we derived and machine-verified an explicit 682-edge witness whose local-field sum $S = 340$ converts under MPR95's Hamiltonian $H/N = -S/(2^8\sqrt{8})$ to their reported improved $D = 8$ bound on the common comparison scale (Section 6.2); we use the same canonical fully frustrated coupling that MPR95 write explicitly (their Eq. 4; derivation documented in Section 6.3); this witness is not claimed to be a reconstruction of any particular configuration MPR may have used internally, since they did not publish one. This 682-edge witness is presented here as the paper's headline lower bound in place of the earlier 680-edge witness.

Writing $g(n)$ for the maximum size of an *odd-square* edge set (every square meets it in exactly one or three edges), the numerical values $g(6) = 132$, $g(7) = 304$, and $g(8) = 682$ are, in physics terms, already implicit in the literature: Derrida, Pomeau, Toulouse, and Vannimenus [5] explicitly construct fully-frustrated ground states for $d \leq 7$ (giving $g(6)$ and $g(7)$ under the correspondence below), and Marinari, Parisi, and Ritort [15] report having obtained, by simulated annealing, a $D = 8$ ground state attaining Derrida et al.'s improved energy lower bound (giving $g(8)$ under the same correspondence). Neither the field identity nor its use to bound $g(n)$ is new: Derrida, Pomeau, Toulouse, and Vannimenus [5] derive the same identity in 1979 by the same argument (a sum over sites in which the frustration of every plaquette cancels the cross terms), note explicitly that it survives restriction to one side of the bipartition, pose exactly the integer optimisation problem Section 6.2 solves, and tabulate its optimum for $n \leq 10$ (their Table I). What Table I lists is the energy bound E_d together with the counts $n(h)$ (its caption describes a ground-state-energy lower bound with the corresponding internal-field distributions; on their scale $E_d = -\bar{h}$, a sign-only relabelling of the mean field, so we write what the table actually prints); the corresponding value $S = 72$ at $n = 6$ is our conversion, not a figure they state, just as the S values we read off MPR95's energies are ours rather than theirs. In their notation the $n = 6$ optimum is realised by what they call the "first solution" ($n(4)$=6; $n(2)$=24; $n(0)$=2), matching our own independently recomputed optimum exactly. The graph-theoretic formulation connecting the fully-frustrated signed hypercube to the C_4-free extremal problem is likewise anticipated by Laplante, Naserasr, Sunny, and Wang [14], who study the frustration index of the fully-frustrated signature directly in these terms (as a graph-theoretic problem they describe, to the best of their knowledge, as previously untreated) (Section 6.2 below). Given this, our contribution at $n = 6, 7, 8$ is narrower than a first reading of Theorem 2 might suggest, and consists of three parts, none of which is the numerical values $72, 160, 340$ themselves: (i) a short self-contained proof (Lemma 12, a mod-8 refinement of the classical two-point inequality) of the optima DPTV tabulate. DPTV formulate the same constrained optimisation over integer distributions, imposing the same parity and sign conditions, and give its best values and the attaining distributions in their Table I; what Lemma 12 adds is a derivation short enough to check by hand, from which those values — and the number of distributions attaining them — follow without the table. It is an alternative route to their results, not a correction or a strengthening of them; (ii) explicit, machine-verifiable edge-list certificates realising the bound at all three of $n = 6, 7, 8$ – DPTV's own realisations are figures and prose for $n \leq 7$ only, $n = 8$ being left open by them and later claimed, without any released data, by Marinari, Parisi, and Ritort [15]; we are not aware of an earlier publicly released explicit odd-square edge list with a dedicated verifier for the $n = 8$ case, nor of an earlier such certificate for the $n = 6$ case. This is deliberately a limited priority statement rather than a claim of exhaustive bibliographic priority: we have not surveyed every literature and data release since 1979; for $n = 6$, for instance, OEIS A245762 lists $a(6) = 132$ with an EXTENSIONS credit to M. Scheucher and P. Tabatabai (2015) for supplying the term, while citing Harborth–Nienborg [11] for the result itself; it links generator scripts, though not an odd-square edge list. Note also that the *values* $g(n)$ for $n \leq 6$, including $g(6) = 132$, are attributed to LNSW26 by [13], so at $n = 6$ what is offered here is a certificate for a known value, not the value; and (iii) an exhaustive decision, in Section 6.3, of a question DPTV posed and left open in 1979 – which of the several $n = 6$ optima described in their Table I are actually realisable by a spin configuration. MPR95 had already addressed this question by simulated annealing, reporting that only the first is realisable and that "the other" seems not to be; we reduce it to a finite (2^{32}) search, decide it in seconds, and thereby upgrade that tentative report to a proof covering *both* non-realisable distributions. Here the prior literature's constructions are not machine-readable in the same sense: DPTV79 give an explicit recursive scaling construction (their bond configurations for H_5, H_6, H_7, described in prose and figures, not released as a computer-readable edge list), while MPR95 report in a single sentence that they exhibited a $D = 8$ ground state matching the improved energy bound, with a method description (simulated annealing) and the bound value (-361 on the D=9 comparison scale they use for bound values across dimensions, not an absolute integer of the D=8 model; see Section 6), but no machine-checkable certificate: no spin configuration, no edge list, and no graph-theoretic translation – an existence claim we can neither independently verify nor extend beyond matching its numerical value. The same holds at $n = 6$, where DPTV79's own $D = 6$ configuration is described in prose; we construct an independent 132-edge odd-square edge list instead, which happens to realise the local-field distribution they report as their table's first solution. Beyond matching that one, Section 6.3 reports a targeted computational search for the other two optimal $n = 6$ distributions DPTV79's Table I lists but do not know to be realisable; extensive simulated annealing (`q6_other_distributions.py`) never reaches either, plateauing at the same nonzero distance from target across every one of 40 and 20 deterministically specified RNG seeds respectively (the consecutive integer seed lists are protocol labels, not a claim of statistically independent random draws), while the same search method reaches the realised distribution in every trial. This is computational evidence, not a proof; Proposition 14 now supplies one (Section 6.3). The exact value $\mathrm{ex}(Q_6, C_4) = 132$ of Harborth and Nienborg [11] is therefore no longer needed to bound $g(6)$ from above, though it remains the source for the unrestricted value. This settles the odd-square subclass completely for $n = 6, 7, 8$ without resolving the unrestricted problem $\mathrm{ex}(Q_n, C_4)$, since a general C_4-free subgraph need not be odd-square.

An exhaustive audit of the 19 866 previously published 304-edge Q_7 solutions shows that only 389 of them lie in the odd-square subclass; the remaining 19 477 do not. The odd-square construction therefore does not subsume or fully explain the structural classification given here (one profile type, Type 18 with 101 solutions, is entirely odd-square; of the other 19, three (ranks 5, 7, and 10) are mixed, containing some odd-square and some non-odd-square solutions, and the remaining 16 contain no odd-square solutions at all): the 19 866 solutions' common structural core (degree sequence $\{4^{32}, 5^{96}\}$, spectral radius in $[4.78543, 4.79129]$, local maximality) and their classification into 20 dimension-profile types are independent results. These 19 866 labelled solutions certainly do *not* exhaust all labelled 304-edge C_4-free subgraphs of Q_7: the first released solution alone has a stabiliser of size 3 in $\mathrm{Aut}(Q_7) \cong (\mathbb{Z}_2)^7 \rtimes S_7$ (order 645 120; exhaustive check over all 645 120 elements), consisting of the identity and two nontrivial elements, both coordinate permutation + XOR pairs. We write a permutation as the tuple p with the meaning "bit i of v becomes bit $p[i]$ of the image", the convention of `type18_automorphisms.py`. Under $v \mapsto \pi(v \oplus m)$ (XOR first, then permute) the two elements are then $(5, 1, 2, 3, 4, 6, 0)$ with mask $m = 33 = 0100001_2$ and its square $(6, 1, 2, 3, 4, 0, 5)$ with $m = 96 = 1100000_2$; under $v \mapsto \pi(v) \oplus m$ the two masks are interchanged. The group itself, and the fact that its order is 3, are of course independent of both conventions; only the labelling of the masks is not. They together form a cyclic group of order 3 – so its orbit alone has size $645\,120/3 = 215\,040 > 19\,866$. Decomposing the whole catalogue (Proposition 20) makes this quantitative: the 19 866 fall into 180 orbits whose lengths sum to 34 227 200, so the released files are 0.058% of the labelled solutions in those orbits alone. What remains open is whether they represent every $\mathrm{Aut}(Q_7)$-orbit of 304-edge C_4-free subgraphs, not whether they exhaust the labelled solutions themselves; see Open Problem 3.

For Q_8, the reconstructed 682-edge odd-square witness is locally maximal in a strong sense: all 342 non-edges of Q_8 create at least 3 new C_4s upon addition (compared with the earlier 680-edge simulated-annealing Solution B, for which 2 of 344 non-edges created only 1 or 2 new C_4s; see Section 5 for both 680-edge solutions released). Every C_4-free-construction bound reported here, and every one of the three odd-square values, is witnessed by an explicit edge-list construction with a machine-verifiable certificate. Two distinct 132-edge Q_6 constructions are released and should not be conflated: `q6_edges_132.jsonl` is C_4-free but not odd-square (its square-intersection histogram is {1:8, 2:44, 3:188}) and establishes

$\mathrm{ex}(Q_6, C_4) \geq 132$, while `q6_odd_square_132.json` is the odd-square witness for $g(6) \geq 132$ (histogram {1:30, 3:210}). For every bound, a complete, deterministic enumeration of all four-cycles (240 for Q_6, 672 for Q_7, 1 792 for Q_8) establishes C_4-freeness, and every *construction* certificate reported here (the edge sets themselves, their C_4-freeness, and the Q_6 and Q_8 odd-square conditions) is reproducible by a dependency-free verifier that is independent of all search and generation code (`verify.py`; *implementation-level* independence is supplied separately by the bundled second implementation `cross_verify.py` described in Section 2), with the Q_7 odd-square membership of individual solutions separately reproducible via `audit_q7_odd_square.py`. The further structural and statistical claims reported below (spectral-radius ranges, pairwise Hamming extrema, Type-18 automorphism counts, non-edge violation distributions, and similar) are largely recomputable from the released code: `analyze_q7_structure.py` reproduces the degree sequence, the 20 direction profiles, the spectral-radius range, and the exhaustive pairwise Hamming statistics (minimum 6, maximum 274, mean 195.396108 . . ., median 196), and `type18_automorphisms.py` reproduces the Type-18 automorphism counts; `reproduce_core.py` drives the certificate and decision checks in one command, and `reproduce_core.py --structural` additionally runs the Q_7 structural and Hamming analyses (which need `numpy` and several minutes). The count of 636 minimum-distance pairs and their breakdown into 28 profile-type combinations are produced by `q7_hamming_tally.py`, which recovers all pairwise distances from a single blocked matrix product (using $|E \triangle E'| = 2(304 - |E \cap E'|)$) in a few seconds and optionally writes the minimum-distance pairs to CSV.

The Q_7 solutions were found by a two-stage process that we have since traced and partly re-verified (the two stages have different evidentiary status, detailed below and in Section 8): a conventional multi-move-type penalty simulated annealing (add/remove/1-swap/kick moves with temperature-scaled Boltzmann acceptance) found a single seed 304-edge solution, and a remove-and-repair collector – starting from an existing solution (optionally transformed by a random element of $\mathrm{Aut}(Q_7)$), removing a few edges, then greedily re-adding only violation-free edges – mass-collected the released 19 866 solutions; we directly confirmed the collector stage by running it ourselves: from the released seed, in one unseeded 30-second run we observed 1 987 new valid solutions. The script sets no random seed, so this exact count is an observed throughput datum rather than a bit-for-bit reproducibility target; separate reruns have shown the same qualitative rapid collection without being used as additional numerical evidence here. Tracing further back, we located most of the prior history, with one correction to an earlier draft of this paper: the earliest recovered attempt (a CBC-solver ILP explicitly commented as attacking Erdős's \$100 problem) has a confirmed bug (its C_4-detection finds 0 of 672 four-cycles, since adjacent vertices in the bipartite Q_7 never share a common neighbour, which its logic incorrectly relies on), so we withdraw our earlier claim that this specific recovered file is what reached 289 edges; a recovered solver log, headed as a corrected version and reporting the correct C_4 count, documents that a since-fixed version of the script did reach 289 edges, but we have not located that corrected script itself. From there, file timestamps together with recovered solver logs and a chat transcript document a progression $289 \to 293 \to 299 \to 301 \to 303 \to 304$ edges, the last step apparently via the `sa_search.py` series. We confirmed the 301-edge step's mechanism directly: `source.py`, the HiGHS version, run for 60 seconds with no prior solution to warm-start from (a genuine cold start), produces a valid, growing 295-edge C_4-free solution via branch-and-bound alone. We further ran a format/implementation-consistency check that any reader can reproduce from the released files alone, without needing our recovered archive: applying `sa_collect304.py`'s own canonicalisation-then-MD5 hash function directly to each of the 19 866 released edge sets reproduces the `hash` field already stored in that same released record, for all 19 866 (Section 8); this shows the released hash field and script are mutually consistent, not that this script historically produced these edge sets, which remains a separate, historical-record claim (recovered logs, chat transcript, timestamps). The two released Q_8 680-edge solutions (Solutions A and B, Section 5) have likewise now been traced to recovered, independently-verified working code, `sa_q8.py`: a penalty-SA hill-climber that always restarts each trial from the current best *valid* (zero-violation) solution rather than from a fresh random sample, progressively improving it (initial greedy construction $\rightarrow$ intermediate saved states at $584, 662, \ldots, 679$ edges $\rightarrow$ 680). We confirmed the two files this recovered code produced – named `q8_edges_680.txt` and `q8_best.json` in the recovered archive, released here as `q8_solution_a.txt` and `q8_solution_b.json` – match Solutions A and B respectively exactly by edge set, and we confirmed the code makes genuine progress on direct execution: run once under an external 90-second timeout from a saved 670-edge intermediate state, it reduced the violation count at the 675-edge target from 22 to 8. `sa_q8.py` sets no random seed, so this specific numeric trajectory is a one-time observation, not a fixed-seed reproducible result: a reader rerunning the released files should expect qualitatively similar monotonic improvement, not these exact numbers. Separate unseeded reruns from the same checkpoint also showed qualitative improvement, but their concrete counts are omitted here because no corresponding logs or fixed seeds are released. This is a different script, released separately from `c4free_sa.py`, whose generalisation of the same two-phase idea to a single n-agnostic script (Section 8) does *not* share this always-restart-from-a-valid-state property, and which we found cannot progress from a fresh random start under its own documented parameters for either Q_7 or Q_8; `c4free_sa.py` is retained as released (with its defect documented) but is no longer the operative account of how either the Q_7 or the Q_8 solutions were produced. For Q_6 we additionally give an ILP formulation whose feasible value matches the known exact value $\mathrm{ex}(Q_6, C_4) = 132$; we did not independently verify the ILP's optimality within a practical runtime. The upper bound itself rests on the combinatorial proof of Harborth and Nienborg [11]; independently, we reproduce the matching lower-bound construction.

Edge lists, ILP files, the odd-square reconstruction and audit scripts, and source code are made available at
https://github.com/minamominamoto/c4free-hypercube . The repository is synchronised with each released revision; readers should check the commit corresponding to this version, since files introduced by a revision appear only from that commit onward – here, among others: `q6_decide_realizability.py`, `solve_field_ip.py`, and the Q_8 second-distribution witnesses. The authoritative file list for this revision, with SHA-256 sums, is given by the three manifests in the accompanying archive: `ORIGINAL_DATA_SHA256SUMS.txt`, `ODDSQUARE_BRIDGE_SHA256SUMS.txt`, and `MANUSCRIPT_SHA256SUMS.txt`; the last covers this document's own `.tex` and `.pdf` so that a reader can confirm which revision a given file set belongs to.

3 C_4 Enumeration and Verification

Lemma 3 (C_4 enumeration in Q_n)**.** *Every four-cycle of Q_n is uniquely determined by a pair of coordinate directions $0 \le i < j \le n-1$ and a base vertex $b \in \{0,1\}^n$ with*

$b_i = b_j = 0$. *Its four vertices are*

$$b, \quad b \oplus e_i, \quad b \oplus e_j, \quad b \oplus e_i \oplus e_j,$$

and the total number of four-cycles in Q_n *is* $\binom{n}{2} \cdot 2^{n-2}$*: there are* $\binom{n}{2}$ *choices of the two varying coordinates and, after forcing those two base bits to* 0*, exactly* 2^{n-2} *choices for the remaining base bits. For* $n = 6$*:* 240*; for* $n = 7$*:* 672*; for* $n = 8$*:* 1 792*.*

Verification algorithm. Given a candidate edge set $E \subseteq E(Q_n)$ stored as a hash set (of unordered pairs $\{u, v\}$, so uv and vu are the same element), the following procedure performs a complete, deterministic C_4-freeness check:

> **for each** pair (i, j) with $0 \le i < j \le n - 1$:
> **for each** $b \in \{0, 1\}^n$ with $b_i = b_j = 0$:
> let $v_1 = b,\ v_2 = b \oplus e_i,\ v_3 = b \oplus e_j,\ v_4 = b \oplus e_i \oplus e_j$
> **if** $\{v_1v_2,\ v_1v_3,\ v_2v_4,\ v_3v_4\} \subseteq E$: **return** VIOLATION
> **return** C4-FREE

We applied this algorithm to all constructions and all 19 866 solutions of Q_7; every call returned C4-FREE.

4 The Q_7 Construction

We constructed a C_4-free subgraph of Q_7 on 304 edges by penalty-based simulated annealing (Section 8, where the production code has since been recovered and verified), then certified it by Lemma 3's algorithm. This construction is released as `selected_edges_best.json`, and its edge set is identical to global index 0 of the 19 866-solution catalogue (first line of `q7_edges_304.jsonl.part1`) – so every property below is directly checkable from either file.

Proposition 4. *The specific* 304*-edge seed construction just described (the first solution found, not a generic member of the later* 19 866*-solution ensemble of Section 7) has the following properties:*

(i) Degree sequence: 32 vertices of degree 4 and 96 of degree 5 (degree sum $608 = 2 \cdot 304$*).*

(ii) Spectral radius $\lambda_1 \approx 4.787$ *(a genuine computed quantity for this construction);* $\lambda_{\min} \approx -4.787$*, i.e.* $\lambda_1 + \lambda_{\min} = 0$ *to machine precision, which is automatic for any subgraph of the bipartite* Q_7 *and serves here only as a sanity check on the eigenvalue computation, not as an independent structural finding.*

(iii) $\mathrm{Tr}(A^4) = 5216$*, attaining the* C_4*-free trace equality* $\sum_i \deg(i)^2 + \sum_{uv \in E}(\deg u + \deg v) - 2|E|$*. Like the item above this is forced by the degree sequence rather than independent of it (the two summands are equal, so the expression is* $2\sum_v \deg(v)^2 - 2|E|$*); see Proposition 19(iii).*

(iv) Local maximality: every non-edge of Q_7 *creates a* C_4 *when added.*

Across the collector runs described in Section 8 (whose `trial` field reaches into the millions per run; 19 866 is the number of *stored solutions*, not the number of trials), we obtained 19 866 C_4-free subgraphs of Q_7 on 304 edges with pairwise distinct edge sets (not quotiented by $\mathrm{Aut}(Q_7)$; this catalogue-level distinctness is asserted mechanically by `verify.py`, which hashes each sorted edge list and requires all 19 866 digests to differ – a separate check from its per-solution duplicate-edge test), all sharing the structural properties stated in Proposition 19.

5 The Q_8 Construction

This section documents the originally announced Q_8 witness, found by a penalty-SA hill-climber distinct from the Q_7 construction's method (see Section 8 for the recovered and verified `sa_q8.py` production code). It has since been superseded as the paper's headline lower bound by the 682-edge odd-square reconstruction of Section 6; it is retained here as an independent structural data point.

Simulated annealing produced the two C_4-free subgraphs of Q_8 on 680 edges (both released in `q8_edges_680.jsonl`), each certified by exhaustive enumeration of all 1 792 four-cycles. We refer to them as Solution A (first line of the file) and Solution B (second line); the two are structurally distinct and are kept separate below, since an earlier draft of this section inadvertently described properties of Solution A and Solution B as if they belonged to a single construction.

Proposition 5. *Both 680-edge solutions are C_4-free and locally maximal (every non-edge of Q_8 creates at least one C_4 upon addition – that is, maximal with respect to edge addition; throughout this paper "locally maximal" means exactly this and implies no stronger optimality notion), with edge density* $680/1024 = 66.4\%$ *(compared with* $304/448 = 67.9\%$ *for the Q_7 construction) and* $\lambda_1 + \lambda_{\min} = 0$ *(automatic for any subgraph of the bipartite Q_8; not by itself a distinguishing structural feature). They differ as follows:*

(i) Solution A: degree sequence $\{4^8, 5^{160}, 6^{88}\}$ *(degree sum* $1360 = 2 \cdot 680$*); square-intersection histogram* $\{1:308,\ 3:1484\}$ *over the* 1 792 *squares (incidentally odd-square, though not constructed as such – see below for the implication of this); non-edge violation distribution* $\{2:3,\ 3:48,\ 4:144,\ 5:136,\ 6:13\}$ *(minimum* 2 *new C_4s per non-edge).*

(ii) Solution B: degree sequence $\{4^7, 5^{162}, 6^{87}\}$ *(degree sum* $1360 = 2 \cdot 680$*); square-intersection histogram* $\{1:302,\ 2:12,\ 3:1478\}$ *(not odd-square:* 12 *squares meet it in exactly* 2 *edges); non-edge violation distribution* $\{1:1,\ 2:1,\ 3:49,\ 4:153,\ 5:124,\ 6:16\}$ *(minimum* 1 *new C_4, attained by exactly one non-edge).*

Solution A's odd-squareness has a notable implication for the timeline of results in this paper: since it was originally announced (before the 682-edge odd-square reconstruction of Section 6 existed) and is itself odd-square, the bound $g(8) \geq 680$ was already achieved within the odd-square subclass at that earlier point, not only via the later, deliberately-constructed 682-edge witness. We did not notice or exploit this at the time; it is recorded here as a fact about the released data, confirmed independently by direct square-by-square enumeration.

Remark 6. Brass–Harborth–Nienborg's tight bound $\frac{1}{2}(n+\sqrt{n}) \cdot 2^{n-1}$ applies only for $n = 4^r$, and their weaker bound $\frac{1}{2}(n + 0.9\sqrt{n}) \cdot 2^{n-1}$ is stated only for $n \geq 9$ (Eq. (1), [3]); neither covers $n = 8$. Evaluating the $n \geq 9$ form at $n = 8$ nonetheless gives ≈ 674.9; both 680-edge constructions and the 682-edge odd-square reconstruction of Section 6 exceed this extrapolated figure (by 5.1 and 7.1 edges respectively), but we do not claim this as a comparison against a proven bound.

5.1 Local optimality of the 680-edge constructions

Solution B is locally maximal in the weakest possible sense among the two: among its 344 non-edges, exactly 1 creates exactly 1 new C_4; exactly 1 creates exactly 2; the remaining

342 create 3 or more (Proposition 5(ii)). Solution A's non-edges all create at least 2 new C_4s.

For historical completeness only, not as a reviewable computational result: the manuscript that originally announced the 680-edge construction also reported a failed search for 681 edges. No seed list or run log survives for that negative-search statistic, and the originally associated conjecture $\mathrm{ex}(Q_8, C_4) = 680$ is false in view of the 682-edge witness. We therefore omit the old numerical trial count from the evidentiary record and retain only the fact that an unarchived failed search was reported, which we mention only as development history, not as evidence bearing on any current claim in this paper.

Table 2 compares the Q_7 construction with Solution B (the 680-edge solution whose local-optimality numbers are discussed above); see Table 3 in Section 6 for the odd-square reconstructions.

Table 2: Structural comparison of the Q_7 construction and Q_8 Solution B (see Proposition 5 for Solution A).

Parameter	Q_7 (304 edges)	Q_8 Solution B (680 edges)
Total edges $\lvert E(Q_n)\rvert$	448	1024
Achieved edges	304	680
Edge density	67.9%	66.4%
Min degree	4	4
Max degree	5	6
Degree sequence	$\{4^{32}, 5^{96}\}$	$\{4^{7}, 5^{162}, 6^{87}\}$
BHN (Eq. 1), extrapolated	≈ 300.2	≈ 674.9

6 The Odd-Square Subclass

6.1 Definition and equivalence to fully frustrated hypercubes

Definition 7. An edge set $E \subseteq E(Q_n)$ is *odd-square* if every square (four-cycle) of Q_n meets E in an odd number of edges, i.e. one or three of its four edges lie in E. Write $g(n) = \max\{\lvert E\rvert : E \text{ is odd-square in } Q_n\}$.

Every odd-square edge set is C_4-free, since no square can meet it in all four edges; hence $g(n) \le \mathrm{ex}(Q_n, C_4)$. The odd-square condition is exactly the condition studied under a different name in the statistical-physics literature on *fully frustrated hypercubes*: assign to each edge e a sign $\sigma(e) = +1$ if $e \in E$ and $\sigma(e) = -1$ otherwise; the square condition becomes "the product of signs around every square is -1", i.e. every square is frustrated.

Lemma 8. *Let $\sigma_1, \sigma_2 : E(Q_n) \to \{\pm 1\}$ both have product -1 on every square. Then there is a function $t : \{0,1\}^n \to \{\pm 1\}$ with $\sigma_2(x, y) = \sigma_1(x, y)\, t(x)t(y)$ for every edge (x, y).*

Proof. Let $\tau = \sigma_1\sigma_2$ (pointwise product). On every square C,

$$\prod_{e\in C} \tau(e) = \Big(\prod_{e\in C} \sigma_1(e)\Big)\Big(\prod_{e\in C} \sigma_2(e)\Big) = (-1)(-1) = +1.$$

The map sending a cycle C to $\prod_{e\in C} \tau(e) \in \{\pm 1\}$ is a group homomorphism from the cycle space of Q_n (over GF(2), under symmetric difference) to $\{\pm 1\}$, since edges shared by two cycles cancel in their symmetric difference and likewise cancel in the product. The squares of Q_n generate its cycle space (a standard fact for hypercubes: every cycle is a symmetric difference of squares – for general n this follows, e.g., from the theorem that every partial cube has a cycle basis consisting of convex cycles [7], the convex cycles of Q_n being exactly its squares; the bundled deterministic script `cycle_space_rank.py` verifies this by direct GF(2) rank computation for every n used in this paper, asserting that the rank of the square incidence vectors equals the cycle-space dimension $E - V + 1 = (n-2)2^{n-1} + 1$ for each of $n = 3, \dots, 8$ – the six dimensions being 5, 17, 49, 129, 321 and 769), so this homomorphism is determined by its values on the squares; since it is $+1$ on every square, it is $+1$ on every cycle. A signature with product $+1$ on every cycle of a connected graph is a *coboundary*: $\tau(x,y) = t(x)t(y)$ for some $t : V \to \{\pm 1\}$ (the classical balance theorem for signed graphs, Harary [8]). Solving for σ_2 gives the claim. □

Lemma 8 says any two odd-square signatures are related by a *vertex switching* t. For a *fixed* coupling J, letting the spin configuration $s : V(Q_n) \to \{\pm 1\}$ range over all 2^{2^n} choices (a sign at each of the $|V(Q_n)| = 2^n$ vertices, hence $2^{|V|} = 2^{2^n}$ assignments) produces exactly the signatures $\sigma_s(x,y) = J(x,y)s(x)s(y)$ in the switching class of J (writing $s = t \cdot s_0$ for a fixed reference s_0 realises every switching t); this is the operational fact we use, not a claim that different s give the same energy under fixed J (they generally do not – that variation is exactly what is being optimised). The energy-preserving gauge transformation in the physical sense instead acts on *both* J and s together, $J(x,y) \mapsto J(x,y)t(x)t(y)$ and $s(x) \mapsto t(x)s(x)$, under which $J(x,y)s(x)s(y)$ is unchanged edge-by-edge [15]. Consequently, $g(n)$ — defined as a maximum over *all* odd-square edge sets — equals the maximum of $|E|$ over the switching class of any one fixed fully-frustrated signature, in particular the reference coupling J used in Section 6.3's witness; optimising that one fixed J (or, for $D = 8$, exhibiting a ground state attaining Derrida et al.'s improved energy lower bound for it, as Marinari–Parisi–Ritort report [15]) therefore does optimise over the class defining $g(n)$, not merely over a possibly-proper subclass of it.

Under the further correspondence between signed hypercubes and Ising spin configurations $s : \{0,1\}^n \to \{\pm 1\}$ via $\sigma(x,y) = J(x,y)\, s(x)s(y)$ for a fixed reference coupling J with every square frustrated, maximising $|E|$ (i.e. maximising the number of *positive* edges, equivalently minimising the number of negative edges) is exactly the problem of minimising the ground-state energy of the fully frustrated Ising hypercube studied by Derrida, Pomeau, Toulouse, and Vannimenus [5] and, for $D = 8$, by Marinari, Parisi, and Ritort [15]: the ground-state energy is (up to an affine rescaling) the number of negative edges, $|E(Q_n)| - |E|$. In the language of signed graphs, $|E(Q_n)| - g(n)$ is the *frustration index* of the fully frustrated signature on Q_n. Laplante, Naserasr, Sunny, and Wang [14] study this frustration index directly (in the context of Huang's proof of the sensitivity conjecture, and as a target they describe as not previously treated in the literature to the best of their knowledge) and explicitly establish a connection between it and Erdős's question on the largest C_4-free subgraph of the hypercube, giving lower and upper bounds on the frustration index of the fully frustrated signature that match when n is a power of 4; our $g(6), g(7), g(8)$ results above sharpen this connection to exact values at $n = 6, 7, 8$ specifically. To quantify that sharpening: our exact frustration indices are $60, 144, 342$ at $n = 6, 7, 8$ (i.e. $|E(Q_n)| - g(n)$). A baseline lower bound LNSW26 state as Theorem 5.2 (which they attribute to an implication of Bialostocki's work [10]) — and which is, in our notation, exactly the Cauchy–Schwarz relaxation $S \le \sqrt{Q\,m} = \sqrt{n}\, 2^{n-1}$ of Lemma 9, already recorded by DPTV79 (p. 620) as giving "a first lower bound for the ground state energy" — $F(H_n, \sigma^*) \geq (n - \sqrt{n}) \cdot 2^{n-2}$, which evaluates to $\geq 57, \geq 140, \geq 331$ at $n = 6, 7, 8$ respectively – looser than our exact values by $3, 4, 11$. LNSW26's own contribution beyond this baseline is a matching upper bound on the frustration index (equivalently, a lower bound on $g(n)$) via a construction that coincides with Brass–Harborth–Nienborg's [3] when n is a power of 4; for $n = 6, 7, 8$ (not powers of 4) this upper bound is strictly looser than the lower bound, so LNSW26's bounds do not determine $g(n)$ at these values. The sharpening therefore applies at $n = 7, 8$ but *not* at $n = 6$: the 20th Emléktábla Workshop problem collection [13] credits LNSW26 with the exact values $g(n) = 1, 3, 9, 24, 56, 132$ for $n \leq 6$. (During source verification we obtained and read the LNSW26 preprint in full, and, on 26 August 2026, the published version's full text (ACM Trans. Comput. Theory **18**(1), Art. 6); the value list appears in neither, so this attribution rests on [13] alone; nor do LNSW26's own published theorems determine $g(6)$, since their upper and lower bounds close only at powers of 4.) The attribution is therefore secondary-source only: *if* it is right (say via a version or communication we have not seen), the record of $g(6) = 132$ predates us in LNSW26; in any case we do not claim the value as new here, and its historical *attainment* is DPTV79's: their $D = 6$ ground state has 60 negative bonds out of 192 (p. 622), i.e. 132 positive edges, and MPR95 note that constructions saturating the improved bound exist for $D \leq 7$. What Lemma 12 adds at $n = 6$ is a short self-contained derivation of a known value; at $n = 7, 8$ it tightens the frustration-index lower bound by 4 and 11, and the explicit witnesses of Section 6.3 close the gap from the construction side, giving exact values where the general bounds leave a nonzero interval. This sign convention matches the one used by the reconstruction script `generate_q8_682.py` (Section 6.3), where E is exactly the set of edges with $J(x, \dim)\, s(x)s(y) = +1$; for a vertex v of degree $\deg_E(v)$ in E, we call $h(v) = 2\deg_E(v) - n$ its *local field* under this convention (matching the local-field histogram reported in Proposition 15 below). We do not reproduce Derrida et al.'s derivation of the local-field lower bound here, and do not need to: Section 6.2 derives the bounds it uses from Lemmas 9 and 12 directly, and cites [5, 15] only for historical priority. Their results are stated there for reference — established for $D \leq 7$ by explicit construction, and used by Marinari–Parisi–Ritort at $D = 8$ — and the corresponding witnesses are constructed and verified independently in Section 6.3. One asymmetry of rigour is acknowledged explicitly: we have *not* machine-verified that DPTV79's own H_6, H_7 bond configurations satisfy the odd-square condition at the certificate level at which our own edge lists are verified. For their constructions the property follows from the correspondence (Lemma 8: the positive-edge set of *any* spin configuration under a fully frustrated coupling is odd-square) applied to their stated construction, not from a data-level check, which is impossible because no machine-readable configuration of theirs was ever published.

This connection was brought to our attention, after the original circulation of this manuscript, by S. Lai (see Acknowledgements), who identified the relevance of [5, 15] and located the improved Q_8 result.

6.2 Exact values for $n = 6, 7, 8$

Derrida et al. [5] give a local-field lower bound on the number of negative (frustrated, in their terminology) edges, tight enough to determine the fully frustrated hypercube's ground-state energy exactly for $D \leq 7$ via an explicit recursive construction, and conjectured it remains tight for $D \geq 8$. For $D = 6$, they give this explicit construction directly: "this configuration for H_6 contains 60 negative bonds (out of a total of 192)" [5, p. 622], i.e.

$192 - 60 = 132$ positive (unfrustrated) edges, with the construction guaranteed fully frustrated (every square meeting it in an odd number of negative edges) by their general recursive method. Under the correspondence of Section 6 this is an explicit odd-square witness with 132 edges, so $g(6) \geq 132$. We cite this for historical priority only: the lower bound $g(6) \geq 132$ is established below by an explicit odd-square edge list of our own (Section 6.3, `q6_odd_square_132.json`), and the matching upper bound $g(6) \leq 132$ follows from Lemma 9 and Lemma 12 without appeal to [11]. We did not reconstruct DPTV's specific $D = 6$ bond configuration, though their recursion determines the edge counts without it.[1] our witness is an independent one, which however turns out to realise the same local-field distribution they report (see Section 6.3).

Lemma 9 (Field identity, after Derrida–Pomeau–Toulouse–Vannimenus)**.** *For any odd-square edge set E in Q_n, writing $U = \{v \in \{0,1\}^n : v$ has an even number of 1-bits$\}$ for the even-parity part of the bipartition of Q_n ($|U| = 2^{n-1}$; the argument below is symmetric in U and its complement, so this choice is only for concreteness and all numerical values reported for U throughout this paper, e.g. Proposition 15's h-distribution, use this same convention) and $h(v) = 2\deg_E(v) - n$ as before, every $h(v)$ with $v \in U$ is an integer of the same parity as n, and*

$$\sum_{v \in U} h(v)^2 = n \cdot 2^{n-1}.$$

Proof. For $v \in U$ and direction $i \in \{0, \dots, n-1\}$ write $s_i(v) = \sigma(v, v \oplus e_i) \in \{\pm 1\}$, so $h(v) = \sum_i s_i(v)$ and $h(v)^2 = n + \sum_{i \neq j} s_i(v) s_j(v)$. Fix $i \neq j$ and consider, for $v \in U$, the square on $\{v, v \oplus e_i, v \oplus e_j, z\}$ where $z = v \oplus e_i \oplus e_j \in U$. Its four edges have signs $s_i(v)$, $s_j(v)$, and (computed from z's own perspective, since a sign is a single scalar attached to the edge) $s_j(z)$, $s_i(z)$. The odd-square condition means their product is -1 — recall the convention fixed at the start of this section: $\sigma(e) = +1$ iff $e \in E$, so a square meeting E in an odd number of edges carries an odd number (4 minus an odd number, i.e. 1 or 3) of -1 factors among its four signs, and its sign product is -1; "odd-square" and "negative square" are the same condition under this convention. Thus: $s_i(v)s_j(v) \cdot s_j(z)s_i(z) = -1$. Writing $X = s_i(v)s_j(v)$ and $Y = s_i(z)s_j(z)$, both lie in $\{\pm 1\}$ and $XY = -1$ forces $X = -Y$, i.e. $s_i(v)s_j(v) + s_i(z)s_j(z) = 0$. The map $v \mapsto z = v \oplus e_i \oplus e_j$ is a fixed-point-free involution on U (as $e_i \oplus e_j \neq 0$), so it partitions U into pairs $\{v, z\}$ on each of which the two corresponding terms cancel; summing over all such pairs gives $\sum_{v \in U} s_i(v)s_j(v) = 0$ for every fixed $i \neq j$. Summing over all ordered pairs $i \neq j$ and adding the $n \cdot |U|$ diagonal contribution gives $\sum_{v \in U} h(v)^2 = n \cdot 2^{n-1}$. □

A scope note before the remark: the odd-square hypothesis enters this proof only square by square, through each individual sign product. The identity itself is exactly

[1] What follows is arithmetic within DPTV's construction, recorded for the reader's convenience; no claim in this paper depends on it. DPTV's construction builds H_d from H_{d-3} by replacing each vertex with a supersite. Counting negative bonds through the recursion gives $12 + 6 + 6 = 24$ at $d = 5$: four H_3 supersites contribute $4 \times 3 = 12$ internal negative bonds, the three white superbonds contribute $3 \times 2 = 6$, and the one black superbond contributes $1 \times 6 = 6$. At $d = 6$, eight H_3 supersites contribute $8 \times 3 = 24$; the minimal H_3 pattern has three black and nine white superbonds, contributing $3 \times 6 = 18$ and $9 \times 2 = 18$, for 60 total — exactly the figure DPTV state in the text. At $d = 7$, the minimal H_4 pattern has 16 supersites and 32 superbonds, eight black and twenty-four white, giving $16 \times 3 + 8 \times 6 + 24 \times 2 = 144$, i.e. $448 - 144 = 304$ positive edges. So the value 304 follows from their construction by a few lines of arithmetic, even though we do not exhibit their H_7 edge set. At $d = 8$ the pattern is H_5 (32 vertices, 80 bonds, minimal negative count 24, matching $r(5) = 56$ positive): 32 supersites and 80 superbonds, 24 black and 56 white, give $32 \times 3 + 24 \times 6 + 56 \times 2 = 352$ negative bonds, i.e. $r(8) = 1024 - 352 = 672$ — the Open Problem 8 table entry whose gap of 10 to the field bound 682 is discussed there.

equivalent to the *average* form of that condition — by the remark below, a side U satisfies $\sum_{v\in U} h(v)^2 = n\cdot 2^{n-1}$ if and only if $\sum_s \mathrm{spl}_U(s) = \binom{n}{2} 2^{n-2}$ — and this average route is how the 19 477 non-odd-square catalogue solutions satisfy it as well, forced by their shared degree sequence (Proposition 19(i)).

Remark 10 (Exact defect formula)**.** The bookkeeping in this proof runs without the odd-square hypothesis and yields an exact formula for arbitrary edge sets. For any $E \subseteq E(Q_n)$, call a square s *U-split at* its even-parity corner v if exactly one of the two edges of s incident to v lies in E (equivalently $s_i(v)s_j(v) = -1$ for the direction pair $\{i, j\}$ of s), and let $\mathrm{spl}_U(s) \in \{0, 1, 2\}$ count the even-parity corners of s at which it is U-split. Each square has exactly two even-parity corners, related by $v \mapsto v \oplus e_i \oplus e_j$, and belongs to exactly one direction pair; running the ordered-pair sum of the proof without the cancellation step, each square contributes $2 - 2\,\mathrm{spl}_U(s)$ once per ordering of its direction pair, so

$$\sum_{v\in U} h(v)^2 = n\cdot 2^{n-1} + n(n-1)\,2^{n-1} - 4\sum_s \mathrm{spl}_U(s) = n^2\,2^{n-1} - 4\sum_s \mathrm{spl}_U(s),$$

the sum running over all $\binom{n}{2} 2^{n-2}$ squares of Q_n. Hence the identity $\sum_{v\in U} h(v)^2 = n\cdot 2^{n-1}$ holds for a given side of the bipartition *if and only if* $\sum_s \mathrm{spl}_U(s) = \binom{n}{2} 2^{n-2}$, i.e. iff squares are U-split at exactly one even-parity corner *on average*. The odd-square condition forces the pointwise version: a sign product -1 over the four edges of s means $s_i(v)s_j(v) = -s_i(z)s_j(z)$ at its two even-parity corners, so $\mathrm{spl}_U(s) = 1$ for *every* s, and symmetrically on the odd-parity side. We use the average-split reformulation diagnostically, not as a search filter: evaluating $\sum_s \mathrm{spl}_U(s)$ already inspects every square, so it is no cheaper than the odd-square test itself. The bundled deterministic script `field_identity_defect.py` verifies the formula – and every numerical instance of it quoted in this paper – on the released certificates and on random edge sets, using only the standard library.

This proof uses only that *every* square has product -1 (the odd-square condition), so it is sufficient for the identity to hold; it is not necessary. Testing against all 19 866 published 304-edge Q_7 solutions – odd-square and non-odd-square alike – shows every one of them satisfies $\sum_{v\in U} h(v)^2 = 448$, including the 19 477 that are not odd-square. This is explained, for these particular solutions, by a simpler fact that does not need odd-squareness either: every one of the 19 866 has all degrees in $\{4, 5\}$ (Proposition 19), and since Q_7 is bipartite, $\sum_{v\in U} \deg(v) = |E| = 304$ exactly (every edge has exactly one endpoint in U); writing a, b for the number of degree-4, degree-5 vertices in U ($a + b = 64$), $4a + 5b = 304$ forces $b = 48, a = 16$ regardless of which specific 304-edge solution is chosen, so $h(v) = 2\deg(v) - 7 \in \{1, 3\}$ takes value 1 on exactly 16 vertices of U and 3 on the other 48, giving $\sum_{v\in U} h(v)^2 = 16\cdot 1^2 + 48\cdot 3^2 = 16 + 432 = 448$ automatically. We verified the U-side split is indeed exactly $\{4^{16}, 5^{48}\}$ for all 19 866, with no exceptions. The identity's failure for the non-odd-square Q_8 Solution B of Section 5 ($1016 \neq 1024$ on U, while the complementary side gives exactly 1024 — so only one side breaks) is therefore not explained by odd-squareness as such: Solution B is locally maximal, like Solution A, but with the smallest possible margin: its non-edge violation multiset is $\{1{:}1, 2{:}1, 3{:}49, 4{:}153, 5{:}124, 6{:}16\}$, so every missing edge still creates at least one C_4, though one of them creates exactly one (against a minimum of 2 for Solution A). It has an asymmetric degree distribution between U and its complement ($\{4^3, 5^{82}, 6^{43}\}$ vs. $\{4^4, 5^{80}, 6^{44}\}$), whereas Solution A, all 19 866 Q_7 solutions, and the 682-edge witness all have symmetric U/complement degree distributions. This symmetry is not itself implied by odd-squareness, however: a small odd-square counterexample in Q_3 (vertices $0, \dots, 7$, $E = \{(0, 2), (2, 6), (3, 7), (4, 5), (4, 6), (6, 7)\}$, whose six squares meet E, in the direction-pair-then-base enumeration order of Lemma 3, in $1, 3, 1, 3, 3, 1$ edges) has U-degree multiset $\{1, 1, 1, 3\}$ against $\{0, 2, 2, 2\}$ on the complement, while still satisfying $\sum_{v\in U} h(v)^2 = \sum_{v\in U^c} h(v)^2 = 12 = 3 \cdot 2^2$ on each side separately. Remark 10 settles the identity's exact logical status for arbitrary edge sets: it holds on a side iff squares are split at one corner of that side *on average*, $\sum_s \mathrm{spl}_U(s) = \binom{n}{2} 2^{n-2}$. Odd-squareness is the pointwise case ($\mathrm{spl}_U(s) = 1$ for every square, both sides at once), and the average can be met without it: the catalogue's very first record – the non-odd-square solution mentioned in Section 6.2 – has per-square split histogram $\{0{:}60, 1{:}552, 2{:}60\}$ on U, sixty deficits exactly cancelled by sixty excesses for a total of $672 = \binom{7}{2} 2^5$, against the pointwise $\{1{:}672\}$ of the odd-square record 34. So the identity is neither a characterisation of odd-squareness (the Q_3 example just given is odd-square without the U/complement symmetry that holds for all 19 866 Q_7 solutions, and record 0 satisfies the identity without being odd-square) nor a coincidence specific to the released catalogue (for the 19 866 it is a forced consequence, via Proposition 19 and $\sum_{v\in U} \deg(v) = |E|$, of *any* 304-edge subgraph of Q_7 having all degrees in $\{4, 5\}$ – a structural fact about that particular edge count, established independently of odd-squareness). The one-sided failure of the Q_8 Solution B noted above is likewise quantified exactly by the formula: on its failing side $\sum_s \mathrm{spl}_U(s) = 1794 = 1792 + 2$, accounting for $\sum_{v\in U} h(v)^2 = 1024 - 4 \cdot 2 = 1016$, while the complementary side has exactly 1792 splits and sum 1024. `field_identity_defect.py` recomputes each of these numbers from the released certificates.

The next lemma sits on no critical path in this paper: no upper- or lower-bound argument below uses it (as re-stated after Lemma 12). We record it because DPTV79 state the restriction and because the released witnesses satisfy it — forcedly, via their degree sequences.

Lemma 11 (Nonnegativity at the maximum)**.** *If E is odd-square in Q_n with $|E|$ maximum (i.e. $|E| = g(n)$), then $h(v) \geq 0$ for every $v \in \{0, 1\}^n$.*

Proof. Suppose $h(v) < 0$ for some v. Flipping the spin at v (replacing $s(v)$ by $-s(v)$, all other spins unchanged) flips $\sigma(v, w)$ for every edge (v, w) incident to v and leaves every other edge's sign unchanged. Every square containing v has exactly two edges incident to v; flipping both leaves their product, and hence the whole square's sign product, unchanged, so odd-squareness is preserved everywhere. At v itself, the positive-edge count changes from $p = (n + h(v))/2$ to $q = (n - h(v))/2$, a change of $q - p = -h(v) > 0$; no edge other than the n edges incident to v changes sign, so $|E|$ increases by exactly $-h(v) > 0$. This contradicts maximality of $|E|$, so no such v exists. □

This is a corollary of, not an alternative to, Lemma 8: the argument uses only that flipping one vertex's spin preserves odd-squareness (immediate from the switching correspondence) and counts the resulting change in $|E|$ directly. All 389 released odd-square Q_7 solutions have 304 edges, which is the maximum by the upper bound $g(7) \leq 304$ proved below in Section 6.2 (the audit locating these 389 solutions is Section 7.4), so the Lemma's hypothesis applies to every one of them; we checked $h(v) \geq 0$ on all 389 with no counterexample — though, like the square split of Proposition 13(ii) and the $\mathrm{Tr}(A^4)$ identity of Proposition 19(iii), this is forced rather than independent: Proposition 19(i) already pins every degree to $\{4, 5\}$, hence $h = 2\deg - 7 \in \{1, 3\}$.

Given Lemma 9 alone, $|E| = (n \cdot 2^{n-1} + \sum_{v\in U} h(v))/2$ is bounded above by the integer optimisation problem of maximising $\sum_{v\in U} h(v)$ subject to the *fixed* sum of squares $\sum_{v\in U} h(v)^2 = n \cdot 2^{n-1}$ over the 2^{n-1} integers $h(v)$ of fixed parity (necessarily $|h(v)| \leq n$, since $h(v) = 2\deg(v) - n$ with $0 \leq \deg(v) \leq n$; the bundled `solve_field_ip.py` imposes that range explicitly, and dropping it changes neither the optima nor the number of optimal distributions) (we use only that every realisable h-vector satisfies these constraints, not the converse that every integer vector satisfying them is realisable as an odd-square signature; the bound obtained is therefore an upper bound on $|E|$, tight whenever a matching witness is exhibited). This is essentially the optimisation problem DPTV79 pose immediately after their own derivation of Lemma 9: maximise $\bar{h}$ (their notation for the mean field) subject to the same sum-of-squares constraint, with one difference: DPTV79 additionally impose $h \geq 0$ before carrying out the distribution optimisation (p. 620, on the stated grounds that a ground-state search cannot have negative field values), whereas we drop that restriction entirely. The restriction is in fact vacuous for every n: replacing h by $|h|$ preserves both the parity condition and $\sum h^2$ while not decreasing $\sum h$, so no optimal vector has a negative entry, and the two problems have the same optimum *and* the same optimal distributions at every n. We none the less state the programme without the sign constraint, since the bound is then visibly independent of any positivity assumption. Their solution method is to observe that the optimal distribution concentrates on the integers nearest $\sqrt{n}$ – unique when exactly two suffice, multiple solutions (their Table I) when three or four are needed, which they note happens at $n = 6$ and $n = 8$ (p. 621) – and to tabulate the resulting optima for $n \leq 10$ by this reasoning rather than by a general closed-form argument. No sign hypothesis enters our derivation: the constraints admit negative h, and Lemma 11 is *not* used. Its role is the separate one of describing configurations that actually attain the maximum (matching what we verified computationally on all 389 odd-square Q_7 witnesses above).

The following elementary observation turns DPTV79's "concentrate near $\sqrt{n}$" heuristic into a closed form that is transparently exact and complete, without appeal to their table. It is the only ingredient beyond the classical two-point convexity inequality they already use.

Lemma 12 (Quantised two-point inequality)**.** *Let a be an integer of the same parity as n and put $b = a + 2$. Then for every integer h of the same parity as n,*

$$(h-a)(h-b) \ \geq \ 0 \qquad \textit{and} \qquad (h-a)(h-b) \equiv 0 \pmod 8.$$

In particular $(h-a)(h-b) \in \{0, 8, 24, 48, \ldots\}$, so it is either 0 (exactly when $h \in \{a, b\}$) or at least 8.

Proof. $h - a$ is even, say $h - a = 2u$ with $u \in \mathbb{Z}$; then $h - b = 2u - 2$, so $(h-a)(h-b) = 4u(u-1)$. Consecutive integers $u, u-1$ have product ≥ 0 and even, so $4u(u-1)$ is a nonnegative multiple of 8. It vanishes exactly when $u \in \{0, 1\}$, i.e. $h \in \{a, b\}$. □

Summing Lemma 12 over $v \in U$ and writing $S = \sum_{v \in U} h(v)$, $m = |U| = 2^{n-1}$, $Q = \sum_{v \in U} h(v)^2 = n \cdot 2^{n-1}$ gives the single inequality used for all three values of n:

$$\Delta \ := \sum_{v \in U} (h(v) - a)(h(v) - b) \ = \ Q - (a+b)S + ab\, m \ \geq \ 0, \qquad \Delta \equiv 0 \pmod 8. \quad (2)$$

Before specialising, one relaxation is worth recording. Cauchy–Schwarz gives $S \leq \sqrt{Q\, m} = \sqrt{n}\, 2^{n-1}$ and hence $|E| \leq 2^{n-2}(n + \sqrt{n})$ — the same expression as the Brass–Harborth–Nienborg lower bound in (1). So at $n = 4^r$ the two meet and $g(n) = 2^{n-2}(n + \sqrt{n})$ exactly. For the lower-bound half of this equality it is not enough that BHN's construction attains that edge count: $g(n)$ requires an *odd-square* set of that size, and this is precisely what

LNSW26 supply — they prove that the BHN construction induces a signature of Q_n in which every 4-cycle is negative (equivalently, that the construction is odd-square), a fact they note does not follow from BHN's original work ([14], Section 5.2 and Theorem 5.4; cf. Section 6 above). With that identification the BHN construction is optimal within the odd-square subclass at those n. This is the shortest explanation of why LNSW26's bounds close precisely at powers of 4, and it locates what Lemma 12 adds: at $n = 6, 7, 8$ the relaxation is not tight. At $n = 7$ nonnegativity alone already gives the exact $S \le 160$, and at $n = 8$ integrality plus the parity of S (each $h(v)$ has the parity of n by Lemma 9, so S is even) gives the exact $S \le 340$; the mod-8 refinement is what is *essential* at $n = 6$, where it excludes $S = 74$. The nonnegativity is the classical bound $S \le (Q + ab\,m)/(a + b)$; the congruence removes whatever residual gap elementary integrality and the parity of S leave between that real bound and the true integer optimum – among our three instances this extra strength is needed only at $n = 6$, as the specialisations below make explicit. No computational filter is invoked in this modular step: Lemma 12 proves term by term that every summand in Δ is a nonnegative multiple of 8, so their sum has the same property. We apply (2) with the pair (a, b) of parity-correct integers straddling the root-mean-square field $\sqrt{Q/m} = \sqrt{n}$, namely $(1, 3)$ for $n = 7$ ($\sqrt{7} \approx 2.65$) and $(2, 4)$ for $n = 6, 8$ ($\sqrt{6} \approx 2.45$, $\sqrt{8} \approx 2.83$); this is the minimising choice, and no search is needed to see it: putting $u = a + 1$ and $Q = nm$, the bound becomes

$$\frac{Q + ab\,m}{a + b} = \frac{m}{2}\Big(u + \frac{n-1}{u}\Big),$$

which is decreasing for $0 < u < \sqrt{n-1}$ and increasing thereafter, so the optimum over real u is at $u = \sqrt{n-1}$; since the admissible u of the correct parity are spaced two apart, it suffices to evaluate the two straddling $\sqrt{n-1}$ and take the smaller. This gives $(2, 4)$ at $n = 6$ ($u = 3$ against $\sqrt{5} \approx 2.24$; the alternative $(0, 2)$ has $u = 1$ and yields 96), $(1, 3)$ at $n = 7$, and $(2, 4)$ at $n = 8$. For completeness, imposing the mod-8 condition on the other adjacent same-parity pairs does not improve these choices: the next-best quantised upper bounds on S are 96, 176, and 408 for $n = 6, 7, 8$ respectively, all weaker than $72, 160, 340$. Pairs with $a + b = 0$ make (2) degenerate and give no bound; $a + b < 0$ reverses the inequality and bounds S from below.

$n = 6$**.** Here $m = 32$, $Q = 192$, $(a, b) = (2, 4)$, so $\Delta = 448 - 6S$. Nonnegativity alone gives only $S \le 74.\overline{6}$, hence $S \le 74$ by parity; but $448 \equiv 0 \pmod 8$ forces $6S \equiv 0 \pmod 8$, i.e. $S \equiv 0 \pmod 4$. The largest such S below $74.\overline{6}$ is

$$S \le 72, \qquad \text{whence} \qquad g(6) \le \frac{192 + 72}{2} = 132.$$

(The exclusion of $S = 74$ just performed is the one place among our three instances where the congruence is essential.) At $S = 72$ we have $\Delta = 16$. By Lemma 12 each vertex contributes a value in $\{0, 8, 24, 48, \dots\}$ (the set $4u(u-1)$, $u \in \mathbb{Z}$); the admissible values not exceeding 16 are 0 and 8 (the next, 24, already exceeds it), and 16 itself is not admissible, so the only way to write 16 as a sum of admissible values is $8 + 8$: exactly two vertices of U have $h \in \{0, 6\}$ and the remaining 30 have $h \in \{2, 4\}$. The witness of Section 6.3 attains $S = 72$ with h-distribution $\{0{:}2,\ 2{:}24,\ 4{:}6\}$ on U, so the bound is tight and $g(6) = 132$ is established without external input — in particular without [11], which we retain only for the unrestricted value $\mathrm{ex}(Q_6, C_4) = 132$ (Section 9).

$n = 7$**.** Here $m = 64$, $Q = 448$, $(a, b) = (1, 3)$, so $\Delta = 640 - 4S \ge 0$ gives $S \le 160$, attained with $\Delta = 0$ (the congruence adds nothing at $n = 7$: it only restates that S is even, automatic since every $h(v)$ is odd and $|U| = 64$ is even). By Lemma 12 equality $\Delta = 0$ forces $h(v) \in \{1, 3\}$ at *every* $v \in U$; combined with $\sum h = 160$ and $|U| = 64$ this pins the distribution to $\{1{:}16,\ 3{:}48\}$ uniquely. Hence

$$g(7) \leq \frac{448 + 160}{2} = 304.$$

Since $S = 2|E| - n\,2^{n-1}$ holds for *any* edge set of Q_n at all, by bipartiteness alone, attaining $S = 160$ at $|E| = 304$ is automatic and does not by itself exhibit an odd-square witness; what is needed is an explicit odd-square set achieving it. The odd-square audit (Section 7.4) supplies 389 such witnesses among the released 304-edge Q_7 solutions; the first in file order is global index 34 (`q7_edges_304.jsonl.part1`, line 35; SHA-256 below is of the canonicalised `edges` array – matching `q7_odd_square_389.csv`'s `solution_sha256` column, not the JSONL line or any `hash` field: `2f2649ae9c4d1c901ba5a88f9825689de3cc19de1b742d86f5d66dbfd3aebec7`), with square-intersection histogram $\{1{:}96,\ 3{:}576\}$, confirming it is genuinely odd-square and attains $S = 160$. (The file's very first record, by contrast, has histogram $\{1{:}36,\ 2{:}120,\ 3{:}516\}$ and is *not* odd-square, despite also satisfying $S = 160$ – exactly because that identity holds regardless of odd-squareness.) So $g(7) = 304$.

$n = 8$. Here $m = 128$, $Q = 1024$, $(a, b) = (2, 4)$, so $\Delta = 2048 - 6S$. Nonnegativity gives $S \leq 341.\overline{3}$; every $h(v)$ is even (parity of $n = 8$), so S is even and

$$S \leq 340, \qquad \text{whence} \qquad g(8) \leq \frac{1024 + 340}{2} = 682.$$

At $S = 340$ we have $\Delta = 8$: *exactly one* vertex of U contributes, and it has $h \in \{0, 6\}$, the other 127 having $h \in \{2, 4\}$. (The mod-8 congruence independently forces $S \equiv 0 \pmod 4$, consistent with the above, but evenness already suffices at $n = 8$; as at $n = 7$, the congruence's essential use is at $n = 6$ only.) Pointwise equality $h(v) \in \{2, 4\}$ everywhere is therefore *impossible* at the optimum for $n = 8$ — in contrast with $n = 7$, where $\Delta = 0$ does force it. The released 682-edge witness attains $S = 340$ with h-distribution $\{0{:}1,\ 2{:}84,\ 4{:}43\}$ on U (independently recomputed from the released edge list), consistent with this description, so the bound is tight and $g(8) = 682$.

We stress that Lemma 12 solves these three instances exactly, not merely up to rounding: an independent exhaustive solution of the integer programme $\max\{\sum h : \sum h^2 = n \cdot 2^{n-1},\ h \equiv n\ (2),\ |h| \leq n\}$ by dynamic programming over the sum-of-squares budget (the displayed $|h| \leq n$ is automatic for any realisable local field because $h = 2\deg_E(v) - n$; it is included only to make the finite computational state space explicit and is not used in the closed-form upper-bound proof) returns optima 72, 160, 340 for $n = 6, 7, 8$ respectively, matching the bounds above, and returns exactly 3, 1 and 2 optimal h-distributions respectively — for $n = 6$, $\{2{:}30, 6{:}2\}$, $\{0{:}1, 2{:}27, 4{:}3, 6{:}1\}$ and $\{0{:}2, 2{:}24, 4{:}6\}$ (labelled A, B and C throughout; in DPTV79's Table I terms, C is their "first solution" and A and B are their "other solutions", the realisable one thus being C, see Section 6.3); for $n = 7$, $\{1{:}16, 3{:}48\}$; for $n = 8$, $\{2{:}87, 4{:}40, 6{:}1\}$ and $\{0{:}1, 2{:}84, 4{:}43\}$. These counts also follow from Lemma 12 in a line, so the enumeration is a check and not a source. The multiplicity phenomenon itself is independently citable: DPTV79 record (p. 621) that when the optimum concentrates on three or four integers they find many solutions, and that this happens exactly at $d = 6$ and $d = 8$ – the two dimensions at which the programme here returns more than one optimal distribution. At $n = 7$, $\Delta = 0$ forces $h \in \{1, 3\}$ pointwise and $\sum h = 160$ over $|U| = 64$ then fixes the multiplicities, giving one distribution. At $n = 6$, the $\Delta = 16$ analysis above already fixes exactly two vertices at

$h \in \{0, 6\}$ and the other 30 at $h \in \{2, 4\}$; writing j for the number of those two at $h = 6$ and k for the number of the remaining 30 at $h = 4$, $\sum h = 72$ reads $6j+4k+2(30-k) = 72$, i.e. $3j + k = 6$, whose solutions $(j, k) = (2, 0), (1, 3), (0, 6)$ give exactly A, B and C. At $n = 8$, the $\Delta = 8$ analysis above leaves one vertex at $h \in \{0, 6\}$ and 127 at $h \in \{2, 4\}$; the same count gives $3j + k = 43$ with $j \in \{0, 1\}$, hence $(j, k) = (1, 40)$ and $(0, 43)$ — the two distributions above. Both $n = 8$ distributions are realised: the released 682-edge witness (`q8_odd_square_682.json`) attains $\{0{:}1, 2{:}84, 4{:}43\}$, and $\{2{:}87, 4{:}40, 6{:}1\}$ is realised by two further witnesses in `q8_B_witnesses.json` (seeds 20261207 and 20261218), independently verified by `verify_q8_B_witnesses.py`: each has 682 edges, square-intersection histogram $\{1{:}301, 3{:}1491\}$ (odd-square), and $\sum_{v\in U} h(v)^2 = 1024 = 8 \cdot 2^7$ (✓). All three $n = 8$ witnesses are distinct edge sets. In fact a single B witness already settles both: its U-side histogram is $\{2{:}87, 4{:}40, 6{:}1\}$ while its U^c-side histogram is $\{0{:}1, 2{:}84, 4{:}43\}$, so one edge set realises each optimal distribution on one side of the bipartition (the 682-edge witness, by contrast, has $\{0{:}1, 2{:}84, 4{:}43\}$ on both sides). Whether some odd-square 682-edge set realises $\{2{:}87, 4{:}40, 6{:}1\}$ on *both* sides we have not determined. For $n = 6$, only distribution C is realised; distributions A and B are provably not realisable (Proposition 14, an exhaustive finite decision). For $n = 7$, the unique optimal distribution is necessarily realised. Thus $n = 8$ is qualitatively different from $n = 6$: *all* optimal field distributions for $n = 8$ admit a realisation, while for $n = 6$ only one does. In each case computation is reproducible with the bundled `solve_field_ip.py` (standard library only, seconds to run); it is a check on the closed-form argument above, not a substitute for it, and the two are logically independent in a sense worth making explicit: the upper bound $S \leq 72, 160, 340$ is established by Lemma 12 alone, via the mod-8 congruence on Δ, without reference to how many configurations attain it – the argument in each of the three cases above shows directly that no integer solution exceeds the stated value, full stop. The DP's role is only to confirm, separately, that the count of optimal h-*distributions* (as opposed to spin configurations realising them) is exactly $3, 1, 2$ – and this too is a complete count, not a sample. To be precise, the code groups equal field values and enumerates count distributions $\{h{:}\text{multiplicity}\}$, not the exponentially many ordered assignments of h-values to vertices; within this count-distribution state space, both the optimisation pass (forward DP) and the enumeration pass (backtracking reconstruction) traverse the same (i, k, s) state tree exhaustively, so completeness of the list follows from the same recursion that establishes correctness of the optimum, rather than from a separate argument.

All three upper bounds $g(6) \leq 132$, $g(7) \leq 304$ and $g(8) \leq 682$ are thus established directly from Lemma 9 and Lemma 12 – reproducing, by a closed-form argument that makes the reasoning self-contained, the $n = 6, 7, 8$ entries among the optima DPTV79's Table I tabulates for $n \leq 10$, and supplying a public machine-verifiable certificate at $n = 8$ after DPTV79 left that dimension open and MPR95 later reported attainment without releasing a configuration (Section 6.3); we retain the citations to Derrida et al. [5] and Marinari–Parisi–Ritort [15] for historical priority and for the connection that prompted this section, not because the upper-bound proof depends on them.

Marinari, Parisi, and Ritort [15] report having exhibited, by simulated annealing, a $D = 8$ ground state attaining Derrida et al.'s improved energy lower bound. Their attainment claim, under the correspondence developed below, corresponds to the same bound value $\sum_{v\in U} h(v) = 340$ established directly above – MPR95 do not themselves report a quantity called S or 340; the correspondence is ours. Their Hamiltonian is $H \equiv -D^{-1/2} \sum_{(x,y)} J_{x,y} s_x s_y$ [15, Eq. 3]; writing $S = \sum_{(x,y)} J_{x,y} s_x s_y$ for the sum over *all* edges (not just U), a bipartite argument identical to the one used for Lemma 9 shows $S = \sum_{v \in U} h(v)$ exactly (every edge has exactly one endpoint in U, so summing the local field over U counts each edge's sign once), so $H = -S/\sqrt{D}$ and, per site, $H/N = -S/(N\sqrt{D})$ with $N = 2^D$. Since each positive edge contributes $+1$ to S and each negative edge contributes -1, $S = |E| - (D2^{D-1} - |E|) = 2|E| - D2^{D-1}$, hence explicitly

$$|E| = \frac{D2^{D-1} - N\sqrt{D}\,(H/N)}{2}.$$

At $D = 8, S = 340$ (equivalently $|E| = (1024 + 340)/2 = 682$) this is, step by step, $H/N = -S/(N\sqrt{D}) = -340/(256\sqrt{8}) = -340/724.077\ldots = -0.46956\ldots$, i.e. $H \approx -120.2$, about -0.470 per site. As a cross-check against MPR95's own text: they report (in their D=9 discussion, pp. 3–4 of arXiv:cond-mat/9410089, the passage checked at page level as recorded in the Acknowledgements) that "on a scale where the minimal energy jump is 1" their D=9 naive bound $H/N = -1/2$ corresponds to an integer value of -384 (i.e. a scale factor of $768 = 384/(1/2)$), and state that on this same scale "the improved lower bound at $D = 8$ gives -361" – note this 768 factor is MPR95's own D=9 comparison scale, not an intrinsic D=8 energy-quantisation unit; they use it purely to compare bound values across dimensions on one common integer axis. Applying the same factor (as MPR95 themselves do for the $D = 8$ figure) to our value gives $768 \times (-0.4696\ldots) = -360.62\ldots \approx -361$, matching their reported integer to within rounding. The identification is not an accident of the rounding width: the neighbouring admissible values $S = 336$ and $S = 344$ give -356 and -365, so among the values $S \equiv 0 \pmod 4$ forced by $\Delta \equiv 0 \pmod 8$, only $S = 340$ rounds to -361 (the intermediate $S = 338, 342$, excluded by that congruence, would give -359 and -363 under the same nearest-integer rounding rule). Three further data points fix and test the scale under the identical rule; their evidentiary roles differ, so we state them once here to keep the count straight: the $D = 9$ *bound* value calibrates the scale (it is what pins 768 down, so it is not an independent check); the $D = 8$ and $D = 10$ values are the two independent checks; and the $D = 9$ *attained* value is a bound-free corroboration at the calibration dimension. First the calibration point: at $D = 9$, $\sqrt{9} = 3$ is itself an integer of the right parity, so the field-bound optimum is the unique uniform distribution $h \equiv 3$, giving $S = 768$, $H/N = -1/2$ exactly, and $768 \times (-1/2) = -384$ – an exact match to the value MPR95 state at $D = 9$. That value is the field-bound target, *not* an attained ground state: MPR95 report failing to reach it, obtaining -0.494792, i.e. -380 on the same scale, four units short. Whether the bound is attained at $n = 9$ is open here too (Open Problem 7). The agreement to be read off this point is therefore between two computations of the same bound, not evidence about $g(9)$. This point, however, is where the scale factor 768 is itself pinned down, so it corroborates the conversion without being independent of it; the two independent checks are $D = 8$ and $D = 10$. At $D = 10$, $m = 512$, $Q = 5120$, $(a, b) = (2, 4)$ gives $\Delta = 9216 - 6S$, so $S \leq 1536$ (with $\Delta = 0$, distribution $\{2{:}256, 4{:}256\}$), and $768 \times (-1536/(1024\sqrt{10})) \approx -364.29 \approx -364$, matching MPR95's reported $D = 10$ value of -364 to the same rounding rule. The scale factor is in fact forced rather than fitted. Every configuration has $S = 2|E| - D\,2^{D-1}$ with $D\,2^{D-1}$ even for $D \geq 2$, so S is an even integer and the achievable values of $H/N = -S/(N\sqrt{D})$ lie on a lattice of pitch $2/(N\sqrt{D})$. (A single spin flip at v changes S by $-2h(v)$ and hence H/N by $2|h(v)|/(N\sqrt{D})$ in magnitude – a vertex-dependent multiple of the pitch, equal to the pitch itself only when $|h(v)| = 1$; for even D every $h(v)$ is even, so no single flip ever realises the pitch there. The quantisation unit is the lattice pitch, not any particular flip.) At $D = 9$ the pitch is $2/(512 \cdot 3) = 1/768$, so 768 is what MPR95's "minimum energy jump equal to 1" scale must mean there; the same pitch at $D = 8$ is $2/(256 \cdot 2\sqrt{2}) = 1/(256\sqrt{2}) \approx 1/362.04$, confirming that 768 is not a $D = 8$ quantisation unit. A further point corroborates it empirically: MPR95 also report their *attained* $D = 9$ value both as -0.494792 and as -380 on the integer scale, and $768 \times (-0.494792) = -379.99\ldots$, so 768 is forced by a datum that involves no bound at all; being a $D = 9$ datum it shares the calibration dimension, so we count it as corroboration rather than as a third independent check. The obvious rival reading — one unit per minimal energy jump in each dimension — is excluded: it reproduces -384 at $D = 9$ but gives $-S/2 = -170$ at $D = 8$, nowhere near MPR95's -361. All four data points – the calibrating $D = 9$ bound, the two independent checks at $D = 8$ and $D = 10$, and the bound-free $D = 9$ attained value – agree under one scale, which is considerably stronger than the $D = 8$ figure alone. Their paper reports the $D = 8$ attainment in energy units only, without an explicit spin configuration or its translation into edge-count terms; we supply that translation and an explicit, machine-verified 682-edge certificate in Section 6.3 below. Under the correspondence of Section 6, this historical attainment report at $D = 8$ corresponds to $g(8) = 682$, matching what was already established above from Lemma 9 together with the integer/parity constraints and elementary inequalities alone (as detailed above, Lemma 11 is not needed for this bound); the explicit witness realising it is due to the present reconstruction, prompted by S. Lai's identification of the connection to [5, 15]. Together with the matching witnesses at $n = 6, 7$ this gives Theorem 2: $g(6) = 132$, $g(7) = 304$, and $g(8) = 682$, each half of each equality now carried by material released with this paper.

6.3 Verification

We constructed explicit witnesses for all three values and machine-verified them independently of [5, 15, 11].

Q_6, $g(6) = 132$. We obtained a 132-edge odd-square witness for Q_6 by the same route used for Q_8 below: the canonical fully frustrated coupling

$$J(x, \dim) = (-1)^{\mathrm{popcount}(x \,\&\, (2^{\dim}-1))},$$

verified fully frustrated on Q_6 (all 240 squares have coupling product -1, no exceptions), together with a 64-bit spin configuration. Because switching preserves frustration, the set of edges on which $J(x, y)s_x s_y = +1$ is odd-square for *every* spin assignment, so the construction reduces to maximising that edge count, i.e. to a ground-state search. We ran a fixed-seed simulated annealing over the 64 spins (`search_q6.py`, seed 20260823), which reached 132 edges on its first trial; the run is exactly reproducible from the released script. The resulting witness is regenerated and self-checked by `generate_q6_132.py` and released as `q6_odd_square_132.json`.

Proposition 13. *The Q_6 witness edge set satisfies:*

(i) *132 edges, all valid Q_6 edges, no repetitions.*

(ii) *Square-intersection histogram $\{1{:}30,\ 3{:}210\}$ over all 240 squares: the odd-square condition holds everywhere, and in particular the set is C_4-free. (Once odd-squareness has independently been verified, and given $|E|$, the split is forced rather than an additional independent check: each edge of Q_n lies in exactly $n-1$ squares, so double counting gives $N_1 + 3N_3 = (n-1)|E|$ and $N_1 + N_3 = \binom{n}{2}2^{n-2}$, whence $N_3 = \big((n-1)|E| - \binom{n}{2}2^{n-2}\big)/2$. This applies equally to the Q_7, Q_8 and Solution A histograms quoted elsewhere in this paper.)*

(iii) *Degree sequence* $\{3^4,\, 4^{48},\, 5^{12}\}$*; degree sum* $264 = 2 \cdot 132$*.*

(iv) *Local-field histogram on* U *equal to* $\{0\!:\!2,\ 2\!:\!24,\ 4\!:\!6\}$*, giving* $\sum_{v\in U} h(v) = 72$ *and* $\sum_{v\in U} h(v)^2 = 192 = 6 \cdot 2^5$*, as required by Lemma 9; the complementary side of the bipartition has the same histogram.*

(v) *Local maximality: no edge of* Q_6 *can be added to it while preserving* C_4*-freeness. More precisely, adding any of the* 60 *non-edges creates at least two new* C_4*s; the distribution of the number of newly completed* C_4*s per non-edge is* $\{2{:}2, 3{:}29, 4{:}26, 5{:}3\}$*.*

The distribution in (iv) is one of the exactly three optimal h-distributions enumerated in Section 6.2, so the witness attains the upper bound $S \le 72$ there and settles $g(6) = 132$. Two further remarks. First, it coincides in multiplicities with the local-field distribution Derrida et al. report for their $D = 6$ configuration, "$n(4) = 6$; $n(2) = 24$; $n(0) = 2$" [5, p. 622]; we did not reconstruct their configuration and do not claim the two edge sets agree, only that our independently found witness realises the same distribution. DPTV79 also report 30 overfrustrated plaquettes out of 240 for their H_6 configuration, matching the 30 squares in the 1-edge class of our histogram $\{1{:}30, 3{:}210\}$ under the complementary positive-edge convention; this is a further consistency check, though a forced one: with 132 positive edges the double count of Proposition 13(ii) already fixes $N_1 = 30$, so this agrees with DPTV rather than confirming anything independent, not an edge-set identity claim. Second, this witness is *not* the released 132-edge Q_6 construction `q6_edges_132.jsonl` of Section 9: their symmetric difference has size $|E \triangle E'| = 62$, and the latter is C_4-free but not odd-square (square-intersection histogram $\{1\!:\!8,\ 2\!:\!44,\ 3\!:\!188\}$, and $\sum_{v\in U} h(v)^2 = 176 \ne 192$). Both are 132-edge C_4-free subgraphs of Q_6; only the new one is a witness for $g(6)$ specifically.

Realizability of the other two optimal $n = 6$ field distributions: an exhaustive decision. The field-bound DP (Section 6.2; `solve_field_ip.py`) identifies exactly three optimal h-distributions on U for $n = 6$, labelled as there: distribution C $= \{0{:}2, 2{:}24, 4{:}6\}$ (realised above), A $= \{2{:}30, 6{:}2\}$, and B $= \{0{:}1, 2{:}27, 4{:}3, 6{:}1\}$. DPTV79 (p. 622) noted that solutions besides their first were listed in Table I and asked whether any is realisable ("We do not know if the other solutions ... are realizable", in the plural; we rely on that sentence for the existence of at least two further solutions, not on a reading of the typeset table itself, which we could not extract). Since DPTV's constraints are the ones solved here and the exhaustive enumeration returns exactly three optimal distributions, their "other solutions" are necessarily A and B; the identification is forced by the completeness of the enumeration rather than by transcription of the table. To be explicit about the evidence: we have read DPTV79 pp. 620–622 directly, but Table I itself does not extract as text from the journal PDF and we have not transcribed it, so every statement here about its contents rests on DPTV's surrounding prose together with our own enumeration. MPR95 reported, by simulated annealing, that at $D = 6$ only configurations corresponding to the first solution of DPTV's Table I are realisable and that "the other solution of the Diophantine equation *seems not to* correspond to any spin configuration" – a tentative negative finding, in the singular, with no released configuration or decision procedure. (Note the grammatical mismatch between the two sources themselves: DPTV79 write "the other solutions" where MPR95 write "the other solution"; on the exhaustive count below – exactly three optima, of which two are non-realisable – the plural is the accurate reading, and we take MPR95's singular as loose phrasing rather than as a competing count.) Since Proposition 13 identifies C with DPTV's first solution by multiplicity, the question is whether A and B are realisable. Here "realisable" prescribes the histogram on the chosen bipartition side U only; U^c is not constrained. This is the same one-side convention used by DPTV79 for $n(h)$. Each of A, B, and C has $S = 72$, so any realisable target automatically has $|E| = (192 + 72)/2 = 132$. We settle the one-side decision exhaustively rather than heuristically.

Proposition 14 (Exhaustive decision)**.** *Among* 132*-edge odd-square subgraphs of* Q_6*, distribution C is realised as the local-field histogram on U – throughout this proposition the histogram is prescribed on the fixed bipartition side U* only*, with* U^c *unconstrained, the same one-side convention as DPTV79's* $n(h)$ *– and distributions A and B are not realised on U by any such subgraph. (By the translation symmetry noted in the proof the restriction to U costs nothing:* $g(n)$ *itself is two-side symmetric, and a distribution realised on one side is realised on the other.)*

Proof. By Lemma 8 the odd-square edge sets of Q_6 are exactly the positive-edge sets arising from spin configurations under *any* fixed fully frustrated coupling, so it suffices to decide the question for the canonical coupling used throughout; the conclusion is independent of that choice. Write $s_v = (-1)^{t_v}$ with $t_v \in \{0, 1\}$, and let $b(v, w) = 1$ exactly when $J(v, w) = -1$; an edge vw is positive iff $t_v \oplus t_w = b(v, w)$. As Q_6 is bipartite, every neighbour of $v \in U$ lies in U^c, so

$$k(v) \;=\; \#\{\, i :\; t_{v\oplus e_i} \oplus b(v, i) = 1 \,\}$$

depends only on the 32 spins on U^c, and $\deg(v) = k(v)$ if $t_v = 1$, $\deg(v) = 6 - k(v)$ if $t_v = 0$. Hence each t_v ($v \in U$) is a free independent choice, and the achievable degree at v is exactly one of $k(v)$, $6 - k(v)$. With $h = 2\deg - 6$, the achievable nonnegative field is $h = 6 - 2m(v)$ where $m(v) = \min(k(v), 6 - k(v)) \in \{0, 1, 2, 3\}$, so $m = 0, 1, 2, 3$ correspond to $h = 6, 4, 2, 0$ respectively. A target h-histogram on U is therefore realisable if and only if the multiset $\{m(v) : v \in U\}$ equals the induced m-multiset for some assignment of the 32 spins on U^c. (This step uses that the target histogram has no negative entry, so that the achievable field at each v is the single value $6 - 2m(v)$; A, B and C are all nonnegative, and by the remark after Lemma 9 no optimal distribution can have one. It also suffices to decide the question on U: translation by e_0 is an automorphism of Q_6 exchanging U and U^c, and it carries edge sets to edge sets preserving both size and odd-squareness, so a distribution realised on one side is realised on the other.) The m-multisets are $\{0{:}2, 2{:}30\}$ for A, $\{0{:}1, 1{:}3, 2{:}27, 3{:}1\}$ for B, and $\{1{:}6, 2{:}24, 3{:}2\}$ for C. This is a finite (2^{32}) decision problem. For A and B it is settled without any search at all, by a parity obstruction. Since $\deg(v)$ is $k(v)$ or $6 - k(v)$, we have $k(v) = (h(v) + 6)/2$ or $(6 - h(v))/2$; these differ by $h(v)$, which is even, so the *parity* of $k(v)$ is determined by $h(v)$ alone, independently of the free spin t_v:

$$k(v) \equiv \tfrac{h(v)+6}{2} \pmod 2, \qquad \text{i.e.} \qquad k(v) \text{ odd} \iff h(v) \in \{0, 4\}.$$

Reducing the definition of $k(v)$ modulo 2 gives $k(v) \equiv \sum_i t_{v\oplus e_i} + \sum_i b(v, i)$, so the parity vector $p \in \mathrm{GF}(2)^U$ is the affine image $p = Mt + B$, where M is the $U \times U^c$ bipartite incidence matrix of Q_6 over GF(2) and $B(v) = \sum_i b(v, i)$. The reachable parity vectors are exactly the coset $B + \mathrm{Im}(M)$. Here M has rank 16, and that coset — 2^{16} vectors — has weight distribution

$$\{8{:}640,\;\; 12{:}13824,\;\; 16{:}36608,\;\; 20{:}13824,\;\; 24{:}640\},$$

so every reachable parity vector has weight in $\{8, 12, 16, 20, 24\}$. A target histogram fixes that weight: it is the number of its vertices with $h \in \{0, 4\}$, namely 0 for A and $1 + 3 = 4$

for B. Neither occurs, so A and B are not realisable — and this needs no enumeration of m-multisets. Distribution C requires weight $2 + 6 = 8$, which does occur, and the witness of Section 6.3 realises it. For completeness we also settled all three by depth-first search with counter-overflow pruning, which agrees (the node counts below are properties of this implementation and its search order, recorded for orientation only: they carry no reproduction obligation, and a differently ordered search will visit different counts; the parity obstruction above is the search-order-independent proof): the search assigns the 32 U^c spins in a fixed index order and keeps one running counter per value $m \in \{0, 1, 2, 3\}$; a $v \in U$ has $m(v)$ determined as soon as the last of its six neighbours in that order is assigned, at which point $m(v)$ is tallied, and the branch is cut the moment any counter exceeds the target's multiplicity for that value. Global spin flip fixes every h, so the first U^c spin may be pinned without loss of generality (each sign configuration has exactly one lift with the pinned value). The search finds a realisation for C (501 nodes visited) and exhausts the space for A (655,359 nodes) and B (22,129,151 nodes) without a realisation; Without the symmetry reduction the two non-realisable searches, which exhaust the space, visit exactly twice as many nodes (1 310 718 and 44 258 302), since the global flip pairs off branches; the realisable case C is not doubled, because the search returns at the first witness found and may find it on either side of the pairing. The released script reports the doubled figure as an expectation for the exhausted cases and recomputes both on request (`--full`). □

The parity obstruction is released as `q6_parity_obstruction.py` (rank, coset weight enumeration and the three targets; standard library only, under a second), and the exhaustive decision as `q6_decide_realizability.py` (also standard library only, about twenty seconds). Its correctness is anchored at both ends: the reduction is checked against the released witness (recomputing m from the spins in `q6_odd_square_132.json` gives $\{1{:}6, 2{:}24, 3{:}2\}$, i.e. C), and the C branch of the search reconstructs a full spin vector whose recomputed h-histogram and edge count are $\{0{:}2, 2{:}24, 4{:}6\}$ and 132. Proposition 14 thus closes the question DPTV79 left open in 1979 and upgrades MPR95's tentative "seems not to" to a proof – for both of the non-realisable distributions, where MPR95 spoke of one (DPTV79's own sentence is plural: "We do not know if the other solutions given in table 1 for $d = 6$ are realizable", p. 622). We stress that the mathematical content of the proposition — of the three optimal distributions, exactly one is realisable on a fixed side — stands independently of any reading of DPTV79's Table I: that reading (obtained, as disclosed in Section 6, from the surrounding prose and the completeness of our own enumeration rather than from a direct transcription) enters only the historical identification of *which* question is thereby closed, not the proposition itself.

Earlier revisions of this manuscript reported instead a heuristic search (simulated annealing on the L_1 distance between the current and target h-histograms; 40 seeds for A and 20 for B at 300,000 iterations each, all plateauing at distance 16 and 8 respectively, against a positive control reaching C in 20/20 runs at 60,000 iterations). What the release contains is not the original runs: the bundled `q6_other_distributions.py` and its CSV/JSON are a re-execution of the same kind of search under a seed protocol newly specified for this package, as the `provenance_note` in both summary files states. They reproduce the numbers earlier revisions reported but are not the historical logs, which were not archived. They are in any case superseded by Proposition 14 and are no longer offered as evidence: a plateau is not a proof, and the asymmetry in iteration budget between the targets and the control makes the phrase "the same search" inexact. The two computations also range over different spaces: the annealing objective allows intermediate configurations with negative local fields (the archived logs record best-histogram entries at $h = -4$), whereas the reduction of Proposition 14 works with the nonnegative field $6 - 2m(v)$ throughout. They agree on A and B, but they are not the same computation restricted.

Q_7, $g(7) = 304$. The 304-edge odd-square value coincides with our originally announced Q_7 lower bound (Theorem 1(i)); of the 19 866 solutions obtained via the SA-seed-plus-collector pipeline (Section 4), 389 are themselves odd-square witnesses of $g(7) = 304$ (Section 7), independently confirming reachability of this value without reference to [5].

Q_8, $g(8) = 682$. We obtained a 682-edge odd-square witness for Q_8 from an explicit 256-bit spin configuration together with the canonical fully frustrated coupling $J(x, \mathrm{dim}) = (-1)^{\mathrm{popcount}(x \,\&\, (2^{\mathrm{dim}}-1))}$ (the coupling MPR95 write explicitly as their Eq. 4; one coupling convention realising a fully frustrated signature; see Section 6 supplementary script `generate_q8_682.py`). The script is dependency-free (Python standard library only), recomputes the edge set from the spin configuration, and asserts every structural quantity below against the recomputed data before writing the certificate; it does not take any of these quantities as input. *Provenance of the spin configuration.* The 256-bit configuration was derived by the author on 17–18 August 2026, using the canonical fully frustrated coupling of Marinari, Parisi, and Ritort [15], as recorded in dated correspondence with S. Lai, who independently reconstructed the identical 682-edge set from the shared spin vector and cross-checked all 1,792 squares, the degree distribution, and the local-field distribution before this material was incorporated into the manuscript. This provenance record is a first-person, dated account (the correspondence itself), not an independently reproducible computational log: no search seed, intermediate search records, or derivation script for the spin configuration itself is included in the released materials, only the scripts that regenerate and verify the edge set *from* the given spin configuration (Section 6). We do not claim this configuration is identical to any specific configuration MPR may have used internally, since they did not publish one; it is an independently derived witness attaining the same bound.

Proposition 15. *The witness edge set satisfies:*

(i) 682 *edges, spanning all* 256 *vertices of* Q_8.

(ii) Square-intersection histogram $\{1\!:\!301,\ 3\!:\!1491\}$ *over all* 1 792 *squares (odd-square condition holds everywhere; zero squares meet the edge set in* 0*,* 2*, or* 4 *edges).*

(iii) Degree sequence $\{4^2,\ 5^{168},\ 6^{86}\}$*; degree sum* $1364 = 2 \cdot 682$.

(iv) Local-field histogram $\{0\!:\!2,\ 2\!:\!168,\ 4\!:\!86\}$ *over all* 256 *vertices, where the local field at a vertex is (positive degree)* $-$ *(negative degree) under the signed-graph correspondence above; restricted to* U *specifically (the* 128 *vertices summed over in Lemma 9 and the upper-bound derivation of Section 6.2), the histogram is* $\{0\!:\!1,\ 2\!:\!84,\ 4\!:\!43\}$*, giving* $\sum_{v\in U} h(v) = 2 \cdot 84 + 4 \cdot 43 = 340$ *directly.*

(v) Local maximality: every one of the 342 *non-edges of* Q_8 *creates at least* 3 *new* C_4*s upon addition (distribution* $\{3\!:\!41,\ 4\!:\!159,\ 5\!:\!120,\ 6\!:\!22\}$*), a strictly stronger local-maximality margin than the originally announced 680-edge Solution B of Section 5, for which 2 of 344 non-edges created only 1 or 2 new* C_4*s.*

All quantities in Proposition 15 were recomputed independently of the reconstruction script by direct enumeration of all 1 792 squares against the released edge list, using the strict definition of C_4-freeness from Lemma 3 (no square may have all four edges present); this independent check found zero violations and is fully reproducible from the released files alone. The reconstruction, its SHA-256 checksum, and an independent audit are additionally reported to have been cross-checked against S. Lai's separately produced implementation, with full agreement; this cross-check is recorded as a dated correspondence account (as in Section 6.3), not as an independently verifiable artefact, since the correspondence, Lai's implementation, and its output are not included in the released materials.

Second $n = 8$ witness. The local-field distribution $\{0{:}1, 2{:}84, 4{:}43\}$ above is one of two optimal field distributions for $n = 8$ returned by the exhaustive integer programme `solve_field_ip.py`. The other, $\{2{:}87, 4{:}40, 6{:}1\}$, is realised by additional witnesses found by the same canonical fully frustrated coupling together with a targeted simulated annealing whose objective is the L_1 distance between the current h-histogram on U and the target $\{2{:}87, 4{:}40, 6{:}1\}$. Over 20 seeds $\times$ 300,000 iterations, 2 runs reached the target ($L_1 = 0$) at seeds 20261207 (iteration 260,621) and 20261218 (iteration 276,842); these seed values are arbitrary RNG integers and are not dates, despite their form. To confirm that nothing depends on that choice, the same search re-run from the unrelated base 90200 also reaches the target (seed 90210, iteration 279 646). The remaining 17 stopped at $L_1 = 6$ and 1 at $L_1 = 8$. Failed intermediate states can contain negative local fields (for example $h = -6$ in one recorded run), because the annealing objective explores non-maximal configurations; Lemma 11 applies only to a maximum odd-square edge set and is not contradicted by such intermediates. Both witnesses are released in `q8_B_witnesses.json` (spins and edges), regenerable via `q8_B_recover.py`, and verified by the dependency-free `verify_q8_B_witnesses.py`. The two successful runs can be checked directly from the stored certificates without any search; the other eighteen are recorded only by their plateau distances, so confirming those requires rerunning the protocol, which `q8_B_recover.py` supports (it is seeded per trial and takes `--seeds` and `--iters`). The 20-run success/plateau tally is provenance information rather than part of the theorem's evidence and is not rerun by the default one-command reproduction. Nothing in the claim depends on it — the assertion is existence, and two certificates carry it:

Proposition 16. *Each witness in* `q8_B_witnesses.json` *satisfies:*

(i) 682 *edges, all valid Q_8 edges, no repetitions.*

(ii) Square histogram $\{1{:}301,\ 3{:}1491\}$ *over all* 1,792 *squares (odd-square; C_4-free).*

(iii) h-histogram $\{2{:}87,\ 4{:}40,\ 6{:}1\}$ *on U,* $\sum h = 340$, $\sum h^2 = 1024$ *(✓). On the opposite side U^c the histogram is* $\{0{:}1, 2{:}84, 4{:}43\}$*, i.e. the other optimal $n = 8$ field distribution. Thus each of these single edge sets realises the two optimal histograms on its two bipartition classes.*

(iv) Local maximality with margin at least 3*. For the witnesses with seeds* 20261207 *and* 20261218*, respectively, the non-edge violation distributions are*

$$\{3{:}34, 4{:}169, 5{:}121, 6{:}18\} \quad \textit{and} \quad \{3{:}41, 4{:}154, 5{:}130, 6{:}17\}.$$

(v) Distinct spin vectors and edge sets, and distinct from `q8_odd_square_682.json`*.*

Thus $n = 8$ differs from $n = 6$: every optimal field distribution for $n = 8$ is realisable by an explicit odd-square 682-edge subgraph of Q_8, while for $n = 6$ only distribution C is

(Proposition 14). The two halves of that contrast rest on different kinds of evidence, and the asymmetry should be stated: at $n = 8$ both realisations are exhibited constructively, whereas at $n = 6$ non-realisability is *proved* exhaustively. Thus "every optimal distribution is realisable" at $n = 8$ means constructive existence for each of the two DP distributions, not an exhaustive decision over all $n = 8$ spin assignments of the kind performed at $n = 6$. We have not attempted the analogous decision at $n = 8$. The obstacle is not the raw size of the space — the reduction gives 2^{127} after pinning one spin, and at $n = 6$ a 2^{32} space was settled in between 501 and 2.2×10^7 nodes — but that the counter-overflow pruning which makes the $n = 6$ search collapse has no guarantee of biting as hard at $n = 8$. The parity obstruction, however, transfers and explains part of the contrast: the rank of the $U \times U^c$ map over GF(2) is $1, 4, 4, 16, 16, 64, 64, 256$ for $n = 2, \ldots, 9$, so it is *full* exactly when n is odd. There is thus no parity obstruction at all at $n = 7$, and one at both $n = 6$ and $n = 8$; at $n = 8$ the criterion reads $k(v)$ odd $\iff h(v) \in \{2, 6\}$, so the two optimal distributions require weights 88 and 84, and the released witnesses show both are attained. This is a second axis, independent of the field identity, along which $n = 7$ differs from its even neighbours.

Table 3 compares the odd-square reconstructions with the SA witnesses from Sections 4 and 5.

Table 3: Odd-square reconstructions compared with the simulated-annealing witnesses of Sections 4–5.

Parameter	Q_7 odd-square	Q_8 odd-square
Edges (value of $g(n)$)	304	682
Source	one of 389 verified SA sols.	explicit certificate at [15]'s bound
Non-odd-square comparison	19 477 catalogue solutions (7.4)	680 (Solution B, 5)
Improvement over that witness	—	+2

7 Structural Classification of Q_7 Solutions

Scope. Every universal statement in this section is over the released catalogue of 19 866 labelled 304-edge solutions unless explicitly stated otherwise. The catalogue is demonstrably not exhaustive of all labelled solutions — the 180 orbits it meets already contain 34 227 200 of them (Proposition 20) — while whether it meets *every* Aut(Q_7)-orbit is unknown, since the total number of orbits is not known (Open Problem 3). Either way, none of the shared structural properties below is claimed for every possible 304-edge C_4-free subgraph of Q_7. This section is an independent contribution about that catalogue: apart from the odd-square audit of Section 7.4 (the 389), its results neither depend on nor feed into Theorems 1 and 2.

7.1 Dimension profiles and the twenty types

Definition 17 (Dimension profile)**.** For $G \subseteq Q_7$, let $e_i(G) = |\{uv \in E(G) : u \oplus v = 2^i\}|$. The *dimension profile* is the sorted tuple $\pi(G) = (e_{\sigma(0)} \geq \cdots \geq e_{\sigma(6)})$.

The dimension profile is invariant under coordinate permutations (a subgroup of $\mathrm{Aut}(Q_7)$ of order $7! = 5040$), giving a coarse but computable classification.

Proposition 18. *Among the* 19 866 *solutions, the dimension profile takes exactly* **20** *distinct values.*

Table 4 lists all 20 types with frequencies and structural invariants.

Table 4: The 20 dimension-profile types occurring among the released 19 866 304-edge C_4-free subgraphs of Q_7, sorted by decreasing frequency. Each profile is the seven per-direction edge counts written in non-increasing order – a canonical form independent of how the seven directions are labelled, which is what makes profiles $\mathrm{Aut}(Q_7)$-invariants and lets the orbit discussion below refer to them directly. H: Shannon entropy of normalised profile (bits).

Rank	Profile π	Solutions	H	λ_1 (type mean)
1	$[44, 44, 44, 44, 43, 43, 42]$	3155	2.8072	4.7869
2	$[45, 45, 45, 43, 43, 42, 41]$	2913	2.8065	4.7880
3	$[45, 45, 45, 43, 43, 43, 40]$	2200	2.8063	4.7870
4	$[44, 44, 44, 44, 44, 43, 41]$	2116	2.8069	4.7875
5	$[46, 46, 46, 42, 42, 42, 40]$	1756	2.8053	4.7874
6	$[46, 46, 46, 42, 42, 41, 41]$	1181	2.8054	4.7887
7	$[44, 44, 44, 44, 44, 44, 40]$	1085	2.8066	4.7868
8	$[45, 45, 45, 43, 42, 42, 42]$	1064	2.8066	4.7877
9	$[45, 45, 44, 43, 43, 43, 41]$	974	2.8067	4.7868
10	$[44, 44, 44, 44, 44, 42, 42]$	931	2.8070	4.7866
11	$[47, 47, 47, 41, 41, 41, 40]$	902	2.8037	4.7880
12	$[45, 45, 43, 43, 43, 43, 42]$	439	2.8069	4.7879
13	$[45, 44, 44, 43, 43, 43, 42]$	307	2.8070	4.7880
14	$[46, 45, 45, 42, 42, 42, 42]$	236	2.8063	4.7891
15	$[44, 44, 44, 43, 43, 43, 43]$	163	2.8073	4.7866
16	$[47, 47, 44, 43, 41, 41, 41]$	153	2.8050	4.7875
17	$[46, 46, 44, 42, 42, 42, 42]$	107	2.8062	4.7888
18	$[48, 48, 48, 40, 40, 40, 40]$	101	2.8014	4.7881
19	$[46, 46, 43, 43, 42, 42, 42]$	44	2.8063	4.7884
20	$[46, 46, 45, 42, 42, 42, 41]$	39	2.8058	4.7878
	Total	**19866**		

Observations: (1) the profile-wise *mean* values of λ_1 shown in the table are rounded independently to four decimal places and lie in $[4.7866, 4.7891]$ across all 20 types (this is distinct from, and narrower than, the exhaustive range $[4.78543, 4.79129]$ over all 19 866 individual solutions reported in Proposition 19 below). (2) H ranges from 2.801 to 2.807, all near $\log_2 7 \approx 2.807$. (3) All 101 Type-18 solutions have a genuine $\mathrm{Aut}(Q_7)$-automorphism (coordinate permutation combined with a bitwise XOR, i.e. the full automorphism group $\mathrm{Aut}(Q_7) \cong (\mathbb{Z}_2)^7 \rtimes S_7$ of Q_7 itself – a proper subgroup of the full affine group $\mathrm{AGL}(7, 2)$ over $\mathbb{F}_2^7$, since only coordinate *permutations* are allowed, not general linear maps – not merely a coordinate permutation) that nontrivially permutes at least two of the three tied 48-edge directions; of these, 46/101 have a genuine order-3 automorphism cyclically permuting all three directions. (An earlier check that considered only pure coordinate permutations, without combining them with the XOR/translation part of $\mathrm{Aut}(Q_7)$, incorrectly found no such automorphisms in a 20-solution sample; that check was in error and has been redone exhaustively over all 101 solutions.)

7.2 Shared structural properties

Proposition 19. *Every solution in the* 19 866 *satisfies:*

(i) Degree sequence $\{4^{32}, 5^{96}\}$.

(ii) λ_1 *ranges over* $[4.78543, 4.79129]$ *across all* 19 866 *solutions (exhaustively computed; not a single shared three-decimal value), with* $\lambda_1 + \lambda_{\min} = 0$ *throughout, automatic for any subgraph of the bipartite* Q_7 *(sanity check, not an independent finding). The type-wise means of Table 4 are reported to four decimal places, ranging from* 4.7866 *to* 4.7891.

(iii) $\mathrm{Tr}(A^4) = 5216$. *(Like the square-intersection split of Proposition 13(ii), this is forced rather than independent: in a* C_4*-free graph any two vertices have at most one common neighbour, so* $\mathrm{Tr}(A^4) = 2\sum_v \deg(v)^2 - 2|E| = 2 \cdot 2912 - 608 = 5216$ *follows from (i) alone.)*

(iv) Local maximality: no edge of Q_7 *can be added without creating a* C_4.

(v) Every edge of Q_7 *appears in at least one solution (verified by taking the union of the edge sets over all* 19 866 *solutions).*

(vi) Every *one of the* 19 866 *solutions admits an automorphism of* Q_7 *of order* 3*; not one has a trivial stabiliser in* $\mathrm{Aut}(Q_7)$.

Item (vi) was first established by the direct check described next; the subsequent full orbit census independently implies it more conceptually and in a sharper form. The direct check is retained because it does not depend on completion of the orbit census. An element of $\mathrm{Aut}(Q_7)$ acts on edges by permuting directions, so it preserves the per-direction edge counts; an order-3 element must therefore have an order-3 coordinate permutation preserving the solution's own direction-count vector (a translation $v \mapsto v \oplus a$ alone has order 2). Grouping the 19 866 solutions by that vector leaves 2 484 classes, and testing the corresponding candidates as column permutations settles all of them. Care is needed with the order condition: for $g(v) = \pi(v) \oplus a$ one has $g^3(v) = \pi^3(v) \oplus (a \oplus \pi(a) \oplus \pi^2(a))$, so g has order 3 exactly when π does *and* $a \oplus \pi(a) \oplus \pi^2(a) = 0$; a candidate failing the second condition has order 6 (its square is then an order-3 element stabilising the same solution, but the check must be stated correctly). With both conditions enforced, every solution is stabilised by at least one order-3 element, with no exceptions (`q7_order3_automorphisms.py`); the 2 484 classes here are the distinct *unsorted* direction-count vectors, a finer partition than the 20 sorted profiles of Table 4. The stabiliser of the first released solution, computed by exhaustive search over all 645 120 group elements, is thus not the special case it appeared to be but an instance of a property shared by the whole released catalogue.

Decomposing the catalogue under the full group sharpens this considerably.

Proposition 20 (Orbit census)**.** *Under* $\mathrm{Aut}(Q_7)$*, of order* 645 120*, the* 19 866 *released solutions fall into exactly* 180 *orbits. The stabiliser orders, one per orbit, are* 3 *(*142 *orbits), 6 (32), 12 (4), 24 (1) and 72 (1); correspondingly the orbit lengths are* 215 040*,* 107 520*,* 53 760*,* 26 880 *and* 8 960*. The orbits meet the catalogue very unevenly: the largest accounts for* 2 048 *solutions (*10.3%*) and the ten largest for* 5 199 *(*26.2%*). The two orbits of stabiliser order* 24 *and* 72 *meet the catalogue in* 55 *and* 46 *solutions respectively, and are exactly the Type-*18 *solutions of Section 7.1: that type is precisely the pair of orbits with the largest stabilisers.*

The census is computed by `q7_orbit_census.py` by exhaustive group action, with no sampling and no extrapolation anywhere. The scan is deterministic: solutions are visited in fixed catalogue order, and to the first still-unassigned solution the script applies *all* $645\,120 = 7! \cdot 2^7$ elements of $\mathrm{Aut}(Q_7)$ (every coordinate permutation composed with every translation), recording which catalogue members appear among the images; a 64-bit random-projection signature with a fixed seed prefilters candidate matches, and an exact comparison of the full 448-entry edge-incidence vector confirms every hit. The exact confirmation is not optional: a single signature collision merges two orbits, and in a signature-only trial run on the odd-square subset it did. Stabiliser orders are then computed per orbit, again exhaustively: only coordinate permutations preserving the representative's direction-count vector can fix it (an automorphism permutes directions), so the scan runs over exactly that subgroup times all 128 translations – a pruning that discards no fixing element – and the orbit length is 645 120 divided by the stabiliser order. Internal cross-checks require the per-orbit catalogue counts to sum to 19 866 and the orbit lengths to 34 227 200. Every step being deterministic, a completed run reproduces the released artefacts byte for byte.

The completed census is released as an artefact, not only as a computation. The file `q7_orbit_census.json` carries the orbit assignment of all 19 866 solutions, the per-orbit catalogue counts, the stabiliser orders and orbit lengths, and the 34 227 200 total, together with the finished checkpoint `q7_orbit_census_ckpt.npz`; both are listed in the SHA-256 manifest. Two verification routes of different strength are bundled, and they should not be conflated. The lightweight `q7_orbit_census_check.py` (standard library, under a second) re-verifies every census-derived number stated in Proposition 20 from the released JSON alone — including, for every orbit, that its length equals $|\mathrm{Aut}(Q_7)|/|\mathrm{Stab}| = 645\,120/|\mathrm{Stab}|$, and that the lengths sum to the labelled total 34 227 200: this checks the *internal consistency of the finished artefact*, not the group action itself, so a reader can audit the stated figures without re-running anything. The strong route is `reproduce_core.py --structural`, which re-runs the census *from scratch*: it invokes `q7_orbit_census.py --fresh`, whose checkpoint and output live in separate `_fresh` files and which never reads the bundled checkpoint, runs it to completion (about six minutes in our single-core container; the budget takes effect only between orbits, so on slower hardware individual orbits can run substantially longer before the first checkpoint – the repository README records the measured environment and times), computes the stabiliser pass on the fresh census, and then requires the resulting `q7_orbit_census_fresh.json` to be SHA-256 byte-identical to the released `q7_orbit_census.json`. (An earlier revision of the driver ran the census in the release directory, where it silently resumed the bundled finished checkpoint, so its "re-run" completed without performing any new search; the `--fresh` mode exists to close that gap.) This revision adds a lighter, certificate-based route that removes the need for any group sweep on the reader's side: `q7_orbit_witnesses.json` records, for every one of the 19 866 solutions, an explicit group element mapping its orbit representative to it, and, for every orbit, the lexicographically minimal incidence vector over all 645 120 images of the representative (the orbit's canonical form) together with an element attaining it. The standard-library checker `q7_orbit_witness_check.py` verifies in well under a minute that every witness element does map its representative to its solution, that the certified assignment coincides with the released census, that each stored canonical form is attained by its stored element, and that the 180 canonical forms are pairwise distinct – certifying the orbit *assignment* (hence "at most 180 orbits") without redoing the sweep. Its `--canonical` mode (numpy; census-scale wall time, sliceable with `--orbits` for time-limited environments) re-verifies that each stored canonical form is the true orbit minimum; since the canonical form is an $\mathrm{Aut}(Q_7)$-invariant, minimality together with pairwise distinctness certifies "exactly 180" independently of the census scan. This answers a review request for an orbit-equivalence certificate lighter than a full recomputation. For calibration of how much the census can carry: the catalogue it decomposes is the de-duplicated output of the recorded heuristic searches; we are not aware of any published 304-edge C_4-free subgraph of Q_7 outside it, and those searches surfaced nothing beyond the 180 orbits — but they were heuristic, so we venture no conjecture about completeness in either direction. The stabiliser pass persists its results into the JSON artefact and refuses an incomplete checkpoint rather than treating the sentinel orbit id -1 as if it were a real orbit.

Three consequences. First, *every* represented orbit has stabiliser order divisible by 3, so Proposition 19(vi), already obtained by a direct all-solution check, also follows from the completed orbit census and holds there in a sharper orbit-level form. Second, summing the orbit lengths gives 34 227 200 labelled 304-edge solutions in the orbits represented here, so the catalogue is 0.058% of even that lower bound — a far stronger statement than the single-orbit argument used earlier, and a quantitative confirmation that the released files are nowhere near exhaustive. Third, the `canonicalize()` defect of Section 8 is not merely a reason for caution: with 180 orbits behind 19 866 labelled solutions and one orbit supplying a tenth of them, duplication under $\mathrm{Aut}(Q_7)$ is demonstrably large-scale, not hypothetical.

This also bears on how item (vi) should be read. The collector's dominant strategies (`auto` on 18 206 records, `auto_seed` on 1 347; the remaining 2 used `repair` and 311 carry no `method` field, see Section 8) act by applying a group element g to an existing solution, and $\mathrm{Stab}(gE) = g\,\mathrm{Stab}(E)\,g^{-1}$, so that operation preserves stabiliser order exactly. But the automorphism is applied to a restart *state*: the collector then deletes k random edges and repairs greedily before storing, so the conjugation argument governs the restart, not the stored solution. What it supports is therefore weaker — the restart preserves stabiliser order exactly, and may thereby bias the subsequent repair toward symmetric regions — rather than a claim that symmetry is propagated to the output. We do not claim this settles the question — it does not explain why the initial seeds were symmetric, and the repair step can in principle break symmetry — but the observation is better read as a plausible artefact of the search lineage than as evidence about 304-edge C_4-free subgraphs of Q_7 in general.

7.3 Solution landscape

Pairwise Hamming distances $d_H(G, G') = |E(G) \triangle E(G')|$, computed exhaustively over all $\binom{19\,866}{2} = 197\,319\,045$ pairs, range from **6** to **274**, with exact mean 195.396108… and exact median 196 (all four values computed directly from the full pairwise distance distribution, not sampled). The minimum, 6, is not attained by a unique pair: it occurs for exactly **636** of the 197 319 045 pairs, spanning 28 distinct profile-type combinations rather than one – most commonly types 3/7 (115 pairs) and 3/5 (87 pairs), with combinations such as 2/4, 2/5, 2/6, 2/10, and 6/8 each occurring dozens of times, down to a single occurrence for one combination (3/20, 1 pair) and two occurrences for three others (12/17, 16/17, and 9/14). Type-1/type-2 pairs, our earlier illustrative example, do occur (22 of the 636) but are far from representative of the minimum's structure. The maximum is rarer still: $d_H = 274$ is attained by exactly **85** of the 197 319 045 pairs. Both extremes are thus vanishingly thin tails around a bulk concentrated near the median 196.

For historical completeness only, not as a reviewable computational result: a failed search targeting 305 edges was also reported. No seed list or run log survives for that negative-search statistic, so we do not retain its old numerical trial count as evidence and no claim in this paper depends on it.

Open exactness question. The data in this paper do not provide a defensible evidentiary basis for a conjecture that $\mathrm{ex}(Q_7, C_4) = 304$: the released catalogue is highly non-exhaustive and the direct SAT/MIP attacks below end in UNKNOWN. We therefore treat exactness at $n = 7$ as an open problem rather than a formal conjecture; see Open Problem 1.

7.4 Odd-square audit

We audited all 19 866 solutions of Section 4 against the odd-square condition of Section 6 (every square meets the solution in exactly one or three edges). Exactly 389 solutions are odd-square; the remaining 19 477 are not. Table 5 gives the dimension-profile breakdown of the 389 odd-square solutions; each belongs to one of only 4 of the 20 profile types of Table 4 (ranks 5, 7, 10, and 18 there).

Most of that restriction is forced, not empirical. Fix directions $i \neq j$. There are 2^{n-2} (i, j)-squares, and each edge in direction i lies in exactly one of them (likewise for j), so $\sum_{(i,j)\text{-squares}} |E \cap C| = e_i + e_j$. Odd-squareness makes each summand odd, and there are 2^{n-2} summands, so for $n \geq 3$ the total is even: $e_i + e_j$ is even for every pair, i.e. *all* e_i *have the same parity*. For $n = 7$ with $\sum_i e_i = 304$ even, that parity must be even. Of the 20 profiles in Table 4, only five (ranks 5, 7, 10, 17, 18) have all direction counts even, so the other fifteen are excluded a priori. (The step from "all e_i share a parity" to "that parity is even" uses $n = 7$: an odd number of odd summands cannot total the even 304. At $n = 6$ both parities remain admissible, so the all-even direction counts $[22, 22, 22, 22, 22, 22]$ of the released Q_6 odd-square witness are an observation, not a consequence.) — and indeed contain no odd-square solution. The genuinely empirical content is therefore the single remaining fact that rank 17 ($[46, 46, 44, 42, 42, 42, 42]$, 107 solutions) has all counts even yet contains no odd-square solution *among the released catalogue*; since the catalogue does not exhaust the labelled solutions, this is not a statement about the type itself.

Since the odd-square subclass accounts for only $389/19\,866 \approx 2.0\%$ of the published solutions, the shared structural core of Proposition 19 and the 20-type classification of Section 7.1 are not fully explained or subsumed by the odd-square construction: the great majority of 304-edge C_4-free subgraphs of Q_7 found here lie outside the fully frustrated switching class of [5]. (These counts are of labelled edge sets, not of $\mathrm{Aut}(Q_7)$ orbits. Odd-squareness is Aut-invariant, so the partition into 389 odd-square and 19 477 non-odd-square is well defined regardless; the ratio, however, does change when measured by orbits. Decomposing the 389 under the full group $\mathrm{Aut}(Q_7)$ of order 645 120 — by applying every group element to each unassigned solution and testing membership on the exact incidence vector — gives exactly **six** orbits, meeting the catalogue in 127, 83, 55, 49, 46

Table 5: Dimension-profile breakdown of the 389 odd-square solutions among the 19 866 published 304-edge Q_7 solutions. Counts in this table are catalogue observations only: absence from a profile class is not a theorem that the corresponding profile can never be odd-square outside the released catalogue.

Rank in Table 4	Dimension profile (sorted)	Count
5	$\{46, 46, 46, 42, 42, 42, 40\}$	78
7	$\{44, 44, 44, 44, 44, 44, 40\}$	127
10	$\{44, 44, 44, 44, 44, 42, 42\}$	83
18	$\{48, 48, 48, 40, 40, 40, 40\}$	101
Total odd-square		389
Total non-odd-square		19 477

and 29 solutions (`q7_oddsquare_orbits.py`). So the 389 labelled odd-square solutions represent only six orbits, and the catalogue is far from a set of orbit representatives. The corresponding decomposition of the whole catalogue is Proposition 20 below: the 19 866 fall into 180 orbits, so the odd-square proportion is $6/180 \approx 3.3\%$ by orbit against $389/19\,866 \approx 2.0\%$ by labelled solution. Membership must be tested exactly: a random-projection signature is faster but a single collision merges two orbits, and in our first run it did, giving 128 and 48 in place of 127 and 49.) The one exception is Type 18 ($[48, 48, 48, 40, 40, 40, 40]$, 101 solutions), all of which are odd-square; the other three types containing odd-square solutions (ranks 5, 7, and 10) are mixed, containing both odd-square and non-odd-square members.

8 Computational Method

The successful outcomes described in this section (the released edge sets themselves) are machine-verifiable certificates: any reader can confirm C_4-freeness and the reported edge counts directly from the supplied data with `verify.py`, independently of how the search was run. The Hamming-distance statistics of Section 7.3 are likewise on a different footing from the search-process claims below: they are computed exhaustively over all released solutions, not sampled, so they are recomputable directly from the released edge lists alone. We release `analyze_q7_structure.py` alongside this revision, which recomputes this section's degree-sequence, dimension-profile, spectral-radius-range, exhaustive Hamming-distance, and Type-18 nontrivial-automorphism-existence claims directly from `q7_edges_304.jsonl.part1--3` in about a single pass over the 19 866 solutions and all $\binom{19\,866}{2} \approx 1.97 \times 10^8$ pairs (peak resident memory around 600–800MB, dominated by the Python object representation of the solution store); running it reproduces the degree-sequence, 20-profile, spectral-radius, and exhaustive Hamming-distance numbers stated for its documented scope. This script confirms only that *some* nontrivial automorphism fixes each Type-18 solution; it does not check the stronger claim made in Section 7 that such an automorphism nontrivially permutes at least two of the three tied 48-edge directions, nor does it determine automorphism *order*: both the direction-permuting property and the 46/101 order-3 figure elsewhere in this paper were established by a separate, more detailed check – now released as `type18_automorphisms.py` (see also `q7_type18_automorphisms_101.csv`) – verifying the automorphism's action on directions,

and $\sigma^3 = \mathrm{id} \neq \sigma, \sigma^2$, directly for each automorphism found.
Update: aspects of the Q_7 production pipeline have been recovered and verified, with important caveats. Earlier versions of this manuscript reported the Q_7 provenance question as unresolved, based on the released `c4free_sa.py` (Section 8.2 below), whose `phase1_sa` we found cannot progress from a fresh random start under its documented parameters. We have since located scripts that produce the 19 866 released Q_7 solutions: `sa_search.py` (a conventional multi-move-type penalty SA – add, remove, 1-swap, and kick moves, with a proper temperature-scaled Boltzmann acceptance rule rather than a hard violation cap) and `sa_collect304.py` (a remove-and-repair collector: starting from an existing valid 304-edge solution – either the seed or a previously found solution, optionally first transformed by a random element of $\mathrm{Aut}(Q_7)$ – it removes a small number k of edges, drawn from $\{1, 2, 3, 4, 5, 8, 12, 20\}$ with fixed weights, then greedily re-adds edges one at a time, accepting only additions that create zero new violating squares, keeping the result if it reaches 304 edges with zero violations and has not been seen before). The caveat: the released `sa_search.py` unconditionally loads its starting state from `selected_edges_best.json` (no random/greedy fallback), and the specific file we recovered and release alongside this revision already contains the final 304-edge solution; running the released script therefore searches for *improvements beyond* 304 edges (consistent with the later, separately-recovered 305-edge search history discussed below), not a from-scratch discovery of the first 304-edge solution. We have since traced the entire earlier chain, with one significant correction to an earlier draft of this section. The earliest recovered attempt on this problem, dated over a week before `selected_edges_301.json` and every other file discussed above, is a CBC-solver (not HiGHS) ILP explicitly commented as attacking "Erdős's $100 problem, the $n = 7$ case" (`source_cbc_origin.py`, released under that name alongside this revision; the script's own internal filename `hypercube_n7.py` reflects its pre-release name). On closer inspection, however, this specific recovered file has a genuine bug: its C_4-detection logic computes, for each edge (u, v), the common neighbours of u and v – but Q_7 is bipartite, so adjacent vertices never share a common neighbour (a shared neighbour of two adjacent vertices would close an odd cycle), and we confirmed by direct execution that this logic finds 0 of the 672 four-cycles, so the ILP it builds has no C_4 constraints at all. A recovered solver log from the same period (`out.log@@20260227183310`, not bundled) is headed with a Japanese label meaning "fixed version" and explicitly prints, in Japanese, "C4 count: 672 – correctly enumerated", confirming a corrected version of the script existed and was run, reaching 289 edges; but we have not been able to locate that corrected script itself among the recovered files, only its log. We therefore withdraw our earlier characterisation of `source_cbc_origin.py` as the script underlying the 289-edge result: it is a recovered early draft with a confirmed defect, retained in the release for transparency about the development history, but it is not the code that produced 289 edges, and no bit-for-bit reproduction of that specific step is available from what we recovered. The codebase was then revised to use HiGHS with warm-starting, and file timestamps together with recovered solver logs document a progression 289 $\to$ 301 (`selected_edges_301.json`) $\to$ 303 (`source_72h.py`'s 72-hour warm-started run, whose log we also recovered and which explicitly records "warm start: loading 301-edge solution" and ends at 303 edges, 0 violations) $\to$ 304 (via the `sa_search.py` series, dated after the 303-edge result). We further located, in a recovered chat transcript documenting the corrected-CBC run above, that it eventually reached 293 edges (after $\sim$ 12 hours) before switching to HiGHS, which reached 299 edges (in 1 hour, independently confirmed 0 violations at the time), refining the chain to 289 $\to$ 293 $\to$ 299 $\to$ 301 $\to$ 303 $\to$ 304. We

distinguish two evidentiary levels here: the 293 and 299 figures are read from a historical chat transcript, not independently re-executed by us (we did not re-run a 12-hour CBC solve or re-verify that specific 299-edge output ourselves); the 289 figure is directly documented by a recovered solver log rather than executed by us either. What we *did* independently execute and verify ourselves: the HiGHS-based `source.py` – the released file hardcodes `time_limit=7200` with no command-line override, so we edited this one value down to 60 for this test – run for 60 seconds with no pre-existing `selected_edges_best.json` (a genuine cold start, not a warm start), finds a valid 295-edge C_4-free solution (0 violations) via HiGHS branch-and-bound alone, confirming this ILP approach genuinely produces growing, valid solutions from nothing and is a plausible, demonstrated mechanism for the 301-edge step. We do not have a bit-for-bit reproduction of the exact original 301-edge solution (that would require rerunning HiGHS for the same duration with the same solver-internal randomisation), but the mechanism at each step is now either independently executed by us, or directly documented by a recovered log or chat transcript, rather than merely claimed, closing the previously-unarchived gap about the search method (though not a bit-for-bit replay of any specific historical run). Turning now to `sa_collect304.py`: unlike `phase1_sa`, this collector never needs to escape a far-from-feasible random state: it always starts already at zero violations and only removes a small number of edges before repairing, so the search stays close to feasible throughout. We verified this directly: the released script hardcodes `RUNTIME=36*3600` with no seed and no command-line override, so we edited the runtime down to 30 seconds for this test; running it from its released seed solution then found 1 987 new, valid, previously-unrecorded 304-edge solutions in that one run. As with `sa_q8.py` above, the absence of a fixed seed means this specific count is a one-time observation of the collector's typical throughput, not a number guaranteed to reproduce exactly on a rerun. We further ran a format/implementation-consistency check, independent of our own recovered archive and reproducible from the released files alone: applying `sa_collect304.py`'s own canonicalisation-then-MD5 hash function, run exactly as implemented, directly to each of the 19 866 released `q7_edges_304.jsonl.*` edge sets exactly reproduces the `hash` field already stored in that same record, for all 19 866 with zero mismatches – without needing the recovered `solutions_304.jsonl` at all. We are careful about what this does and does not show: it demonstrates that the released `hash` field is internally consistent with the released `sa_collect304.py` implementation applied to the released edge sets – a statement about format compatibility between two things we release together – not that this script historically generated these edge sets. The historical-provenance evidence for that (the recovered logs, chat transcript, and file timestamps discussed throughout this section) is separate in kind, not strengthened by this hash check; we keep the two distinct rather than treating the format-consistency result as if it were additional provenance evidence. (This 12-hex-digit MD5 `hash` field is unrelated to the full SHA-256 `solution_sha256` column reported by `audit_q7_odd_square.py` and used elsewhere in this paper, e.g. for the global-index-34 witness above; the two are different hash schemes over the same underlying edge sets, not to be confused with each other.) We must correct our own earlier description of what this function does, however: we had described it as coordinate relabelling by descending per-direction edge count, followed by taking the lexicographically smallest of the 2^n bit-flip images. On rereading the code, this describes what a docstring-level intent might have been, not what is implemented. The implementation instead loops over only the n single-bit XOR masks (not all 2^n combined masks), and compares candidate frozensets with Python's `<` operator, which for `frozenset` tests *proper subset*, not lexicographic order; since every candidate here has the same cardinality (304), no candidate is ever a proper subset of another, so this comparison is always `False` and the flip-selection loop never updates its choice. The function's actual output is therefore just the coordinate-permuted set from the first step, with no flip-based canonicalisation occurring at all. We verified this discrepancy directly: recomputing the canonical form as originally (mis)described – trying all $2^n = 128$ XOR masks with a genuine lexicographic comparison – gives a different result from the released implementation for the first published solution (the true lexicographic minimum is attained at XOR mask 107, not reproduced by the actual code), and for 92 of the first 100 released solutions, this correctly-described alternative hash does not match the `hash` field actually stored. There is a second, independent gap in this function's canonicalisation beyond the flip-loop: the coordinate-permutation step sorts directions by descending edge count, but ties are broken only by Python's stable sort (original label order), not by any canonical rule; two graphs in the same $\mathrm{Aut}(Q_7)$-orbit that differ only by permuting two equal-count directions can therefore receive different hashes. We confirmed this directly: the first released solution has direction edge-counts $[44, 43, 44, 42, 43, 44, 44]$ (four tied at 44), hash `79014e2f4aed`; swapping directions 0 and 2, two of the four tied at 44 (an automorphism of Q_7), changes the computed hash to `0ba97c019106`. The other five transpositions of tied directions give five further distinct values, so the instability is systematic rather than an isolated collision. The 19 866/19 866 match we report above is against the code exactly as implemented (with this defect and the flip-loop defect above), which is what we release and what actually produced the stored `hash` values; it is not evidence that full translation-class *or* coordinate-permutation canonicalisation is occurring, and the released 19 866 solutions should not be assumed de-duplicated across the full automorphism group $\mathrm{Aut}(Q_7)$ on this basis (only that they are 19 866 pairwise distinct *labelled* edge sets, as already stated in Proposition 19). We do not know whether the intended, fully-canonicalising version was ever the one actually run in production, or whether this defect was present throughout; we did not attempt to guess or silently correct it, for the same reason given throughout this section. This also fully explains the previously undocumented per-record metadata fields (Section 8 above, as reported in earlier manuscript versions), independently of this defect, since the fields' meanings come from the surrounding collector logic, not from `canonicalize()` itself: `method` records which of the collector's three restart strategies produced that solution (`auto`: random automorphism of a random existing solution; `repair`: direct remove-and-repair of a random existing solution, no automorphism; `auto_seed`: random automorphism of the original seed); the 311 records lacking a `method` field and the 8 carrying a `run` rather than a `trial` field come from an earlier version of the collector script (which we examined during recovery as `sa_collect304 (0).py`, but which we have not included in the released package – it is not among the files a reader receives) that logged records with only a `run` counter and no `method` distinction, before the script was revised to its released form; `trial` is the sequential attempt counter within a single collector run, and `elapsed` the wall-clock seconds since that run started (consistent with a shell history showing the collector was run and re-started under `nohup` multiple times, explaining why `trial` resets are visible across the released file). We also note a genuine implementation defect in the recovered `sa_search.py` itself: `cur_list`/`mis_list` (the candidate lists sampled for removal/swap moves) are rebuilt from `current_set`/`current_missing` only every `REBUILD_INTERVAL`= 5 000 iterations, while ordinary add/remove moves update `current_set`/`current_missing` directly every iteration without refreshing these lists; a stale list entry can therefore be chosen for removal or swap, and since `delta_remove` does not check that its argument edge is actually present, the tracked `current_v` can in principle drift from the true violation count. `count_violations` is called once, to initialise `current_v` at the start of each run, but is not called again to re-verify before `save_best` writes an improved solution to disk. This is a defect in the search code's internal self-consistency, not in the released certificates: every released 304-edge solution – regardless of which script produced it – has been independently re-verified C_4-free by exhaustive enumeration in this manuscript (Section 3), so the defect does not affect the correctness of $\mathrm{ex}(Q_7, C_4) \geq 304$. We release the code with this defect documented, not silently patched, for the same reason given throughout this section: patching it would misrepresent what was actually run. We release `sa_search.py`, `sa_collect304.py`, and the seed solution alongside this revision as recovered code for the Q_7 results, with the two caveats above (the seed's provenance below 304 edges, and this list-staleness defect); see Section 8.2 below for the separate, distinct `c4free_sa.py` whose relationship to the Q_8 results (and to the Q_7 results, before this recovery) remains as previously described. The old numerical trial counts for the 305-edge and 681-edge failed searches are unaffected by this recovery and remain unsupported by surviving seed lists or logs; we therefore omit them from the evidentiary record. Only the 680-edge *solutions* themselves (not the negative-search statistics) have since also been traced to verified code, described next.

Reproduction safety for recovered search scripts. The recovered production-history scripts write fixed filenames in their current working directory and should be executed only in a disposable working copy, not in the release directory. In particular, `source.py` and `sa_search.py` overwrite `selected_edges_best.json`; `sa_collect304.py` appends `solutions_304.jsonl` and can also overwrite `selected_edges_best.json` if it reaches 305 edges; and `sa_q8.py` writes/overwrites `q8_best.json`. `source.py` additionally writes `selected_edges_<k>.txt` for the edge count k it reaches. The two search-recovery scripts `q8_A_recover.py` and `q8_B_recover.py` default their output to `q8_A_witness_run.json` and `q8_B_witnesses_run.json` respectively — deliberately *not* to the manifest-listed `q8_A_witness.json` and `q8_B_witnesses.json`, which an earlier revision did overwrite on a bare run. Some verification/analysis scripts also create new side files (for example `q6_spins.txt` and the orbit-census checkpoint/JSON). The SHA-256 manifests authenticate only the files they explicitly list; the appearance of additional generated files is normal, whereas overwriting a listed seed file changes the authenticated release.

8.1 Update: aspects of the Q_8 680-edge production pipeline have also been recovered

We have since also located and verified the actual production code for the two released Q_8 680-edge solutions: `sa_q8.py`, a penalty-SA hill-climber structurally similar to `c4free_sa.py`'s Phase-1/Phase-2 design (and sharing its per-trial hard violation cap, `max_viol`, drawn from $[8, 35]$) but critically different in one respect: each trial restarts not from a fresh random sample but from `best_valid_sol`, the current best *zero-violation* solution (initially an edge-greedy construction, then whatever the algorithm has since improved upon), loaded from `q8_best.json`. Because every restart point already has zero violations, the freezing failure mode of `c4free_sa.py` does not arise: proposals only need to stay within `max_viol` of an already-feasible state, not close an initial gap of 100+ violations. The recovered directory contains a progression of intermediate saved solutions (584, 662, 666, 669, 670, 672–680 edges), consistent with gradual hill-climbing improvement over many trials. We verified two things directly: first, that the recovered `q8_edges_680.txt` and `q8_best.json` match the released Solution A and Solution B (Section 5) exactly by edge set (we release these two files as `q8_solution_a.txt`/`q8_solution_b.json` alongside this revision, so a reader can repeat this edge-set comparison directly); second, by direct execution – resuming from the 670-edge intermediate state under an external 90-second wall-clock cutoff (`timeout 90 python3 sa_q8.py`, not the script's own `RUNTIME` variable) – that the code makes genuine progress, reducing the violation count at the 675-edge target from 22 to 8 within that window. We release this 670-edge checkpoint as `q8_checkpoint_670.json`; to repeat this second test, copy it to `q8_best.json` in the same directory as `sa_q8.py` before running (the script only autoloads a file named exactly `q8_best.json`), and use an external timeout as above rather than editing the internal `RUNTIME` variable: `RUNTIME` is checked only once per outer trial, while each trial's inner Phase-1 loop runs a fixed 2–12 million steps with no time check inside it, so setting `RUNTIME` to a small value does not reliably stop execution near that value – only an external, process-level timeout does. `sa_q8.py` sets no random seed, so a rerun will show the same qualitative monotonic-improvement behaviour (confirming the code is not frozen, unlike `phase1_sa`) but not necessarily the same specific violation-count trajectory or the exact $22 \to 8$ figures – this is a one-time observation of genuine progress, not a fixed-seed reproducible numeric result; without this file, `sa_q8.py` instead starts from a fresh greedy construction, which is a different (and also legitimate, since the algorithm always restarts from a valid state either way) starting point than the one we tested from. We also release `sa_q8.py` itself. A historical aggregate count for failed 681-edge attempts was recorded in earlier manuscript versions, but we did not recover the corresponding seeds or log; the count is therefore omitted here rather than repeated as an unverifiable statistic.

8.2 The released `c4free_sa.py` and its known defect

We have since determined why `c4free_sa.py` does not match the production history: it was written after the fact. Following an early external review (of the first arXiv submission, which did not yet include search source code) that flagged the absence of published search code as the submission's principal weakness, `c4free_sa.py` was written on 2026-03-28 – after the Q_7 (19 866 solutions, complete by 2026-03-17) and Q_8 (680-edge solutions, complete by 2026-03-21) results already existed via the separate scripts recovered and described above – specifically to have *some* search code available for release, and was validated only on the much smaller Q_5 case (56 edges) at the time, not at the Q_7/Q_8 scale where the `viol_cap` defect documented below becomes fatal. This explains, rather than excuses, the discrepancy: `c4free_sa.py` was never the operative research code for either dimension; it was a retrospective, small-scale-validated addition made to satisfy a reviewer's reproducibility request, and the defect went undetected because it was not exercised at the scale that triggers it. The earlier quantitative claim about the Q_8 *negative-search process* remains unarchived, as noted just above, and its numerical trial count is omitted from this revision. Before the two recoveries above, the provenance gap described the Q_7 and the Q_8 680-edge solution claims as well. The search program `c4free_sa.py`, distinct from both recovered production-code pairs above, is released alongside this manuscript as originally reported, but the specific run seeds, logs, and driver invocations that would substantiate the old 681-edge negative-search statistic are not supplied, and we no longer believe `c4free_sa.py` as released is what produced either the Q_7 or the Q_8 solutions (see the recoveries above for what we now believe did). This process-level claim should be read as the authors' record of the search that was run, not as an independently reproducible certificate in the same sense as the edge sets. Worse, as detailed below, we found on direct execution that the released, unmodified `c4free_sa.py` frequently cannot progress at all from a fresh random start under the documented parameter ranges, for a structural reason (not merely slow convergence); we could not identify parameters under which it reaches the target edge count with zero violations. A reader wishing to verify the exact 681-edge process-level statistics should therefore not assume rerunning `c4free_sa.py` as documented will reproduce them, or indeed complete a single trial successfully; see below for the specifics.

Beyond the aggregate counts, each released Q_7 record itself carries the provenance fields (`method`, `run`, `trial`, `hash`, `elapsed`) explained above. Of the 19 866 records, `method` is `auto` for 18 206, `auto_seed` for 1 347, `repair` for 2, and absent for 311 — so among the collector's three restart strategies represented in the released catalogue the third contributed essentially nothing, a point worth keeping in mind when reading the description of the three as parallel options. The implementation also uses the label `seed` for its initial seed record; no released catalogue row carries that label, so it is not a fourth restart strategy in the metadata summary above; a `run` field is present on only 8 records (one with `run=0`, seven with `run=2`); and 19 858 records carry a `trial` field with values up to at least seven figures. These fields are now explained by the recovered `sa_search.py/sa_collect304.py` pipeline above, subject to the two caveats noted there (the seed's own provenance below 304 edges, and the list-staleness defect); this record is substantially more resolved than the analogous Q_8 681-edge negative-search provenance claim, for which no run log or seed list survives (the Q_8 680-edge *solutions* themselves, by contrast, have also now been traced to recovered code – see below).

8.3 Two-phase simulated annealing

Phase 1 (Penalty SA). The acceptance criterion for proposed moves minimises $f(E) = -|E| + \lambda V$ where V is the number of C_4 violations and $\lambda \in [0.30, 0.90]$, but the "best" state `phase1_sa` tracks and returns is *not* the minimiser of f: the released code instead keeps the state with lexicographically smallest $(V, -|E|)$ seen so far (`v_cur < best_v or (v_cur == best_v and size_cur > best_size)`) – prioritising fewer violations first, and only using edge count as a tiebreaker among states with equal V. Temperature schedule: geometric from $T_0 \in [0.20, 4.00]$ to $T_1 \in [0.001, 0.030]$ over 3–15 million steps. A move toggles a single edge; the change in f is computed incrementally in $O(\deg)$ time using precomputed C_4-to-edge incidence lists.

Phase 2 (Swap SA). Edge count is fixed; swap moves (remove one edge, add one non-edge) minimise V. Phase 2 only runs when phase 1 *returns* a best-so-far state with $V \le 4$ violations – `phase1_sa` tracks and returns the best state seen during the run, not necessarily its terminal current state, so a literal reimplementation using only the terminal state could behave differently. In the released code, the step budget for phase 2 is chosen by the phase-1 violation count v, not by dimension: 2×10^8 steps if $v \le 1$, otherwise 5×10^7 steps. In practice this correlates with dimension, since phase 1 typically ends closer to $v \le 1$ at the Q_7/305-edge target than at the Q_8/681-edge target, but the branch itself is on v, not on n or the target edge count.

Diversification. The released search code (`c4free_sa.py`) computes, before each trial, a random element of $\mathrm{Aut}(Q_n)$ (random bit permutation composed with random bit flip) applied to the current best solution, intended as a diversified restart point. However, this computed value is not passed into `phase1_sa`, which unconditionally initialises from a fresh uniform-random edge set of the target size (`rng.sample(range(ne), target)`) regardless of any prior best solution; the automorphism-based restart is dead code in the released implementation. (The script's own docstring originally described this mechanism without noting the bug; we have appended a note to the docstring documenting it, without altering any functional code, so that a reader of the code alone is not misled.) The same applies to two further recovered scripts: `sa_search.py` and `sa_q8.py` each carry a comment marked `NOTE (added retrospectively)` — in `sa_search.py` recording that the seed load is unconditional, in `sa_q8.py` recording that the Brass–Harborth–Nienborg figure the script prints as a "lower bound" at start-up is an extrapolation outside the cited domain and not a proven bound at $n = 8$ (Section 8.5). These three files — `c4free_sa.py`, `sa_search.py`, `sa_q8.py` — are the complete list of released scripts carrying annotations we added after recovery. The literal marker `NOTE (added retrospectively)` occurs in the latter two; `c4free_sa.py` records its retrospective findings under `KNOWN BUG` comments instead. In each case only comments or docstrings were touched, and the SHA-256 sums in `ORIGINAL_DATA_SHA256SUMS.txt` are those of the annotated files, not of the historical originals.

A more serious defect in `phase1_sa`. On direct execution, we found that the released `phase1_sa` is frequently unable to make *any* progress at all, for a structural reason independent of step count. Each accepted or rejected proposal toggles a single edge, changing the violation count V by at most $n - 1$ (the number of squares containing that edge); a proposal is rejected outright whenever the resulting V would exceed `viol_cap` (drawn per trial from $[6, 40]$). If the initial random edge set's V exceeds `viol_cap` $+ (n - 1)$, *no* single-edge toggle can ever produce an accepted move, so the state is frozen from step one, for any number of steps. We verified this is not a rare occurrence but the typical case: reproducing the exact RNG state of trial 1 for `--n 7 --target 304 --seed 42`, the initial violation count is $V = 152$ against `viol_cap` $= 14$ (best single-toggle result: 147, still far above cap), and running `phase1_sa` for 100 000 steps leaves $V = 152$ unchanged.[2] Reproducing trial 1 of `--n 8 --target 680 --seed 0` similarly gives initial $V = 333$ against `viol_cap` $= 38$, frozen after 50 000 steps. Sampling 10 000 independent uniform-random edge sets directly (bypassing the annealing loop; `random.Random(999)`, `rng.sample(range(ne), target)` repeated 10 000 times per dimension, minimum V tracked via `count_violations`) gives a minimum V of 117 for $Q_7/304$ and 297 for $Q_8/680$ – both far above the maximum possible `viol_cap` of 40 – so this is not attributable to unlucky seeds; independent resampling with other seeds gives minima in the same range (we observed 111–117 for Q_7 and 296–304 for Q_8 across several seeds), so the qualitative conclusion (minimum $V \gg 40$) does not depend on the particular seed chosen, though the exact minimum does. We could not identify parameter draws under which the released, unmodified `phase1_sa` reaches $V = 0$ from a fresh random start for these targets. Consequently, our earlier claim that fresh per-trial random initialisation explains the diversity of the released solutions should be read with this caveat: we can confirm the released *solutions* are valid (independently verified C_4-free certificates), but the released, unmodified `phase1_sa` – run as literally documented – cannot be what produced them, for the structural reason just demonstrated (not merely bad luck with parameters). We now know why: as detailed in Section 8.2 above, `c4free_sa.py` was itself written after the fact

[2] To reproduce these RNG states, note that `search()` draws its five per-trial parameters and *then* performs one further, unused draw `start = rng.sample(range(ne), target)` that is never passed to `phase1_sa` (this is the dead-code diversification restart described above). That draw must be consumed before generating the initial edge set; skipping it gives $V = 129$ instead of 152.

(dated 2026-03-28), once the Q_7 and Q_8 results already existed via the separately-recovered scripts described above, and validated only at the much smaller Q_5 scale at the time – it was never a candidate for the historical production code in the first place, so there is no remaining question of whether some undocumented earlier version of *this script* matches a different historical parameter setting. The genuine remaining gaps are narrower and specific: the very first sub-304-edge bootstrap step of `sa_search.py`'s own lineage (Section 8 above), and the unarchived 681-edge negative-search statistic (Section 5), neither of which this section's finding about `phase1_sa` bears on directly.

The diversity actually observed across the 19 866 collected solutions is therefore not fully explained by the mechanisms described in this section; we note for completeness that even had the intended automorphism-based restart mechanism been wired correctly, applying an automorphism to a solution does not change which $\mathrm{Aut}(Q_n)$-orbit it belongs to, and the annealing moves used here (uniform edge toggle/swap proposals against an automorphism-invariant objective) admit an $\mathrm{Aut}(Q_n)$-equivariant Markov kernel; any diversification benefit would have come only from decorrelating the specific labelled trajectory taken by a finite stochastic run; it does not diversify the orbit sampled from.

8.4 Comparison with known lower bounds

Brass–Harborth–Nienborg's weaker bound is stated for $n \geq 9$; the values below evaluate that formula at $n = 7, 8$ as an extrapolation for illustrative comparison, not as a proven bound at those dimensions (see Table 1 and the Remark in Section 5).

$$n = 7: \quad \tfrac{1}{2}(7 + 0.9\sqrt{7}) \cdot 64 \approx 300.2 \text{ (extrapolated)}, \quad \text{bound: } 304\ (\Delta = +3.8,\ +1.27\%),$$
$$n = 8: \quad \tfrac{1}{2}(8 + 0.9\sqrt{8}) \cdot 128 \approx 674.9 \text{ (extrapolated)}, \quad \text{bound: } 682\ (\Delta = +7.1,\ +1.05\%).$$

The $n = 8$ figure is the odd-square reconstruction of Section 6; the simulated-annealing method of this section independently reaches 680 edges on Q_8 ($\Delta = +5.1$, $+0.75\%$; Section 5).

8.5 Regularity trend across dimensions

Degree sequences across dimensions: Q_6: $\{4^{56}, 5^8\}$; Q_7: $\{4^{32}, 5^{96}\}$; Q_8: $\{4^8, 5^{160}, 6^{88}\}$ (Solution A; Section 5). The minimum degree is 4 in all three of these C_4-free witnesses; the degree support is $\{4, 5\}$ for both Q_6 and Q_7 (unchanged from $n = 6$ to $n = 7$) and widens to $\{4, 5, 6\}$ only at Q_8. Note, however, that the odd-square Q_6 witness of Section 6.3 (Proposition 13) has degree sequence $\{3^4, 4^{48}, 5^{12}\}$ with minimum degree 3, and is nevertheless locally maximal; thus minimum degree 4 is *not* a necessary feature of 132-edge C_4-free subgraphs of Q_6, only a feature of the particular witness `q6_edges_132.jsonl` reported here. Beyond this, the exponents do not follow an evident closed-form pattern: Q_6's $\{56, 8\}$ split is markedly more skewed than Q_7's $\{32, 96\}$ or Q_8's $\{8, 160, 88\}$, so we do not claim a general regularity trend here.

9 Computational Reproduction of the $\mathrm{ex}(Q_6, C_4) = 132$ Lower Bound

Harborth and Nienborg [11] proved $\mathrm{ex}(Q_6, C_4) = 132$ by combinatorial arguments. We reproduce the lower-bound (construction) side fully independently and computationally

(Section 9.1); for the upper-bound side we give an ILP formulation whose feasible value matches 132 (Section 9.2), but we did not independently verify its optimality within a practical runtime, so the upper bound $\mathrm{ex}(Q_6, C_4) \leq 132$ itself rests on [11], not on our own solve.

9.1 Lower bound: explicit construction

The operative machine-verified witness for $\mathrm{ex}(Q_6, C_4) \geq 132$ in this revision is the odd-square certificate `q6_odd_square_132.json` (odd-square implies C_4-free), whose generation is fully traced: a fixed-seed search (`search_q6.py`, seed 20260823) over the canonical fully frustrated coupling, regenerated and self-checked by `generate_q6_132.py` (Section 6.3). A second, historically earlier 132-edge construction is retained as auxiliary data, used by no theorem of this revision: simulated annealing (Section 8) is reported to have found a C_4-free subgraph of Q_6 on 132 edges, certified by exhaustive enumeration of all 240 four-cycles. That explicit edge list is provided in `q6_edges_132.jsonl` and is kept, under continued verification, for the record. Unlike the Q_7 and Q_8 cases, we have not recovered a production script or historical run log that traces the particular file `q6_edges_132.jsonl` back to its generating search. The certificate itself is independently valid, but its search provenance remains unresolved. The released, unmodified `phase1_sa` nevertheless exhibits the same structural freezing at $n = 6$, target 132: reproducing trial 1 for `--n 6 --target 132 --seed 42` gives initial $V = 54$ against `viol_cap` $= 14$ (best single-toggle result 49, still above cap), frozen after 100 000 steps; `--seed 0` gives initial $V = 52$ against `viol_cap` $= 38$ (best single-toggle result 48). A reader tracing these numbers must consume the RNG in the order the code does: `search()` first samples a diversification start and *discards it unused* (the defect of Section 8.2), and `phase1_sa` then draws a second sample from the same stream – it is this second sample whose initial V is quoted above, and that value is internal to `phase1_sa` (the script prints only the final violation count). Following the stream without consuming the discarded sample gives $V = 56$ (seed 42) and $V = 58$ (seed 0) instead; the discrepancy is itself the most direct numeric exhibit of the discarded-start defect. The 132-edge certificate itself remains independently verified as a valid C_4-free construction. In contrast to the partially recovered Q_7/Q_8 lineages, however, no recovered production code or historical log currently connects this particular Q_6 edge set to a generating run, so no such provenance claim is made.

9.2 Upper bound: integer linear program

A framing remark first: this subsection documents a deliberately retained *unfinished* computational experiment. No claim in this paper rests on it — the upper bound $\mathrm{ex}(Q_6, C_4) \leq 132$ rests on [11] throughout — and the formulation is kept in full for two reasons: it is the specification of the released artefacts `q6_ilp.mps` and `q6_ilp_edge_map.csv`, which a reader may wish to attack with a stronger solver, and it is the substance of the withdrawal, recorded on the title page, of v1–v4's claim of a computational proof.

Label the 192 edges of Q_6 as $x_0, \ldots, x_{191} \in \{0, 1\}$ (matching the released `q6_ilp.mps`'s 0-indexed variable names; the text elsewhere in this paper is otherwise 1-indexed for readability, but the MPS file itself uses 0-indexing throughout), and let $\mathcal{C}_6$ denote the set of all 240 four-cycles of Q_6. For each four-cycle $\{e_1, e_2, e_3, e_4\} \in \mathcal{C}_6$, the constraint

$x_{e_1} + x_{e_2} + x_{e_3} + x_{e_4} \le 3$ ensures C_4-freeness. The ILP is:

$$\max \sum_{i=0}^{191} x_i \quad \text{subject to} \quad \sum_{e \in C} x_e \le 3 \ \ \forall C \in \mathcal{C}_6, \quad x_i \in \{0, 1\}.$$

This formulation has 192 binary variables and 240 constraints, each variable appearing in exactly 5 of the 240 square constraints (matching Q_6's edge-square incidence). The MPS file `q6_ilp.mps` is provided for independent verification (SHA-256 in the released checksum manifest). The correspondence between each x_i and its Q_6 edge was not originally recorded alongside the MPS file. We have since constructed an incidence-preserving identification (`q6_ilp_edge_map.csv`: $x_i \leftrightarrow$ vertex pair) by matching the MPS's variable–constraint incidence pattern against Q_6's known edge–square incidence structure under one specific, standard enumeration convention (integer vertex labels $0, \ldots, 63$; squares enumerated by base vertex, then direction pair). We call this an identification rather than *the* original correspondence because incidence structure alone determines it only up to $\mathrm{Aut}(Q_6)$: any automorphism compatible with the assumed labelling convention would give an equally valid, incidence-matching relabelling. The identification is nonetheless useful for interpreting individual solution vectors and is consistent with (not merely coincidentally matching) the released data throughout. The checkable content is stronger than that caution might suggest, and we state it as a testable claim: under this identification the 240 four-element variable sets of the MPS constraints coincide with the 240 squares of Q_6 *exactly, as a family of sets* – not merely incidence-compatibly – and any reader can test this mechanically from `q6_ilp.mps` and `q6_ilp_edge_map.csv` alone.

Proposition 21. *The ILP above has feasible value* 132*, matching the released 132-edge construction (Section 9.1); combined with Harborth and Nienborg's independent combinatorial proof that* $\mathrm{ex}(Q_6, C_4) \le 132$ *[11], this gives* $\mathrm{ex}(Q_6, C_4) = 132$*. We did not independently verify optimality of* 132 *for this ILP instance within a practical runtime (see below); the upper bound* $\mathrm{ex}(Q_6, C_4) \le 132$ *used here rests on [11], not on our own solve of the ILP. We are, moreover, not aware of a published independent re-verification of that combinatorial proof; readers for whom this matters should weigh the unrestricted* $n = 6$ *equality accordingly (the odd-square value* $g(6) = 132$ *is unaffected, being self-contained here).*

We independently re-solved `q6_ilp.mps` with HiGHS 1.15.1 (Python bindings via `highspy`; default settings, single thread, single-threaded x86_64 Linux VM, Intel Xeon @ 2.10GHz, 4GB RAM). A feasible solution of value 132, matching the released construction's edge count but not necessarily its edge set (the symmetry of Q_6 admits many optima), was found quickly, but the solver did not close the optimality gap: under a 240-second limit the run ended at primal -132, dual bound -140 (a 6.06% gap), after 60 146 branch-and-bound nodes – the dual bound still eight units from the primal. As this figure is implementation- and machine-dependent, it should be read as evidence that the gap does not close quickly under a generic default-settings solve, not as a precise benchmark. This is consistent with the large automorphism group of Q_6 ($|\mathrm{Aut}(Q_6)| = 6! \cdot 2^6 = 46\,080$) producing many symmetric alternative optima, which is known to slow generic branch-and-bound without dedicated symmetry-breaking. We do not claim generic MIP solvers verify this instance's optimality quickly; the 132-edge value should be regarded as independently established by Harborth–Nienborg's combinatorial proof [11], with the ILP providing a machine-checkable feasible witness for the primal side and a formulation that a sufficiently long or specialised solve can, in principle, verify on the dual side.

10 Open Problems

1. Prove or disprove $\mathrm{ex}(Q_7, C_4) = 304$. A direct SAT-encoding attack on this question (a 305-edge C_4-free existence query encoded as CNF, 64 512 variables and 129 104 clauses, with a symmetry-broken variant at 113 664 variables/227 786 clauses; attempted cube-and-conquer splitting via `march_cu` and direct solving via CaDiCaL and Kissat) was attempted over several weeks and did not reach a conclusion: the cube-generation step itself repeatedly failed to complete, and direct solver runs returned `UNKNOWN` rather than a SAT/UNSAT verdict. A second, independent attempt using a different encoding (Sinz's sequential at-most-143 cardinality encoding on the negated edge variables, arriving at the same 64 512-variable, 129 104-clause scale by a different construction) and solving directly with Kissat (no cube splitting) likewise terminated with no SAT/UNSAT verdict logged. A third, independent approach – direct branch-and-bound on the original 448-variable ILP with added symmetry-breaking constraints (685 rows total), using HiGHS and warm-started from the known 304-edge solution – did not close the optimality gap, which remained at 5.26% (best bound ≈ -320.0 against the incumbent -304) when the log ends. Each run was allowed to continue for days rather than hours; we do not quote wall-clock figures, since they are hardware-dependent and unfalsifiable for a reader. The instance sizes and outcome figures above are *reported results*, not re-generable artefacts of this release: the CNF files and their encoders, the symmetry-broken MIP formulation (685 rows), the solver command lines and versions, and the final logs are not bundled, so a reader can neither regenerate nor audit them from the released materials. We record all three as unsuccessful attempts, not as evidence either way; the only conclusion they support is that the *specific formulations and configurations we tried* did not settle the instance – encoding choices, presolve and symmetry-breaking options, and solver versions all remain unexplored dimensions, so we draw no generalisation about off-the-shelf SAT or MIP solvers as a class. A speculative structural remark on where a 305-edge example would have to live, if one exists. Since $g(7) = 304$, any such example is necessarily non-odd-square, and Remark 10 quantifies how far from odd-square it must be: for any 305-edge set, $\sum_{v \in U} \deg(v) = 305$ on *each* side of the bipartition, and an exhaustive dynamic programme over integer degree sequences shows the minimum of $\sum_{v \in U} h(v)^2$ is then 456, attained only by the degree multiset $\{4^{15}, 5^{49}\}$; since $456 > 448$, the field identity fails strictly on *both* sides, so each side carries a split deficit of at least two squares, $\sum_s \mathrm{spl}_U(s) \le 670 = \binom{7}{2} 2^5 - 2$. Separately, every 305-edge C_4-free set contains 305 distinct 304-edge C_4-free subsets, each extendable (restore the deleted edge) and hence *not* locally maximal, whereas all 19 866 released solutions are locally maximal (Proposition 19(iv)); so none of those 305 subsets coincides with any catalogue member, and a 305-edge example, if one exists, remains invisible to the released catalogue even after deleting any single edge. Neither observation is evidence of nonexistence; each only pins down where a counterexample would have to live: at least two split-defective squares away from odd-square on each side, and strictly outside the catalogue's single-edge-deletion neighbourhood.

2. Determine $\mathrm{ex}(Q_8, C_4)$ exactly. Is it 682, or can a non-odd-square construction exceed $g(8) = 682$? The Q_7 audit of Section 7.4 shows the two questions are genuinely different for $n = 7$ (the released 304-edge solutions are overwhelmingly non-odd-square), so no assumption is made here about which way $n = 8$ resolves. Our

current understanding of the structural side, stated as such: the parity obstruction of Proposition 14 operates *inside* the odd-square class – it constrains which field histograms a fully frustrated spin configuration can realise – and says nothing about non-odd-square subgraphs; we know of no structural obstruction to a non-odd-square C_4-free subgraph exceeding $g(n)$. What the data show is only this: at $n = 6$, where ex is known, nothing exceeds $g(6)$; at $n = 7$ every known 304-edge solution matches $g(7)$ while 98% of them lie outside the subclass. We read the latter as weak evidence that the odd-square constraint is not what pins the known extremal count, not as evidence about whether $g(8)$ can be exceeded.

3. What is the number of $\mathrm{Aut}(Q_7)$-orbits among all 304-edge C_4-free subgraphs of Q_7? The released catalogue meets exactly 180 of them, six of which contain the odd-square solutions (Proposition 20); what is unknown is how many orbits exist altogether, and hence whether the 19 866 meet every one. Since those 180 orbits already contain 34 227 200 labelled solutions, a complete enumeration by labelled solution is out of reach, but an orbit-level enumeration may not be.

4. Is the degree sequence $\{4^{32}, 5^{96}\}$ necessary for any locally maximal C_4-free subgraph of Q_7 on 304 edges?

5. Can the flag algebra method of [1, 2] yield tight bounds for small $n \leq 8$?

6. Does the degree support of locally maximal C_4-free subgraphs of Q_n near the extremal edge count widen beyond $\{4, 5\}$ again for $n \geq 9$ (as it did once, from Q_7 to Q_8; Section 8.5), and is minimum degree 4 necessary at these edge counts for $n \geq 9$? (At $n = 6$ it is already not necessary; see Section 8.5.) The Q_9–Q_{15} certificates of [12] are explicit data bearing on this, though we have not tabulated their degree sequences here.

7. The field bound makes $n = 9$ the simplest case beyond $n = 8$: $\sqrt{9} = 3$ is a parity-correct integer, so the optimal h-distribution is the uniform $h \equiv 3$ and the upper bound $g(9) \leq \frac{2304+768}{2} = 1536$ follows without Lemma 12. MPR95 explicitly report failing to find a $D = 9$ ground state by simulated annealing. We have not applied the fully frustrated search of Section 6.3 to $n = 9$; whether the field bound is tight there ($g(9) = 1536$?) is unknown. The *unrestricted* problem at $n \geq 9$ is no longer without explicit data: R. Wrona has released C_4-free edge-list certificates for Q_9–Q_{15} in the Erdős Problems discussion thread for problem 86 [12], obtained by product lifts along hypercube automorphisms followed by local ILP repair, with Q_9 at 1 505 edges and Q_{11} at 7 164; the author of the present paper carried out an independent certificate-level check of that set by sparse cherry counting (see the disclosure below). Those are lower bounds on $\mathrm{ex}(Q_n, C_4)$, not odd-square witnesses, and no exactness is claimed for them; but $1\,505 \leq \mathrm{ex}(Q_9, C_4)$ sits only 31 below the field bound $g(9) \leq 1536$, so the two questions are close at $n = 9$. Whether any $n \geq 9$ certificate is odd-square we have not determined.

8. **(Damásdi; also, in the equivalent frustration-index form, Z. Wang.)** This question is not ours: it is Problem 1 of "Odd squares in the hypercube" in the 20th Emléktábla Workshop problem collection [13], posed there in the same notation ($f(n) = \mathrm{ex}(Q_n, C_4)$, $g(n)$ the odd-square maximum) and with the same one-line argument for $g(n) \leq f(n)$; the collection also records that Z. Wang posed it in

frustration-index form. We restate it because Theorem 2 bears on it directly, not as a new problem. The equality $g(n) = \mathrm{ex}(Q_n, C_4)$ already holds for the small exact cases $n = 4, 5, 6$: the known unrestricted values are $24, 56, 132$ (Table 1). The pattern is worth tabulating, since it extends further down and breaks in an informative place. Writing $r(n)$ for the positive-edge count of DPTV's recursion — H_d built from H_{d-3} with $|V(H_{d-3})|$ supersites at three negative bonds each, r's own negative bonds as black superbonds at six, and the rest white at two — and $g(n)$ for the field bound of Section 6.2:

n	2	3	4	5	6	7	8
DPTV construction $r(n)$	3^*	9^*	24^*	56	132	304	672
field bound on $g(n)$	3	9	24	56	132	304	**682**
known $\mathrm{ex}(Q_n, C_4)$	3	9	24	56	132	—	—

Starred entries are DPTV's base constructions (their Figs. 1–3), not outputs of the recursion, which they state from $d \geq 5$. The three agree wherever all three are known, and the recursion meets the field bound exactly up to $n = 7$ and then falls short by 10 at $n = 8$. That shortfall is the quantitative content of DPTV's own remark that for $d \geq 8$ their construction gives an energy strictly above their bound: it is why 682 needed a search rather than the recursion, and it marks the first n at which the two notions separate. For $n = 4, 5$, the matching odd-square lower bounds are taken from DPTV79's explicit fully frustrated constructions (no $n = 4, 5$ machine-readable witnesses are bundled here), while for $n = 6$ the present package contains its own independently verified odd-square certificate. For $n = 7, 8$, Theorem 2 only establishes $g(7) = 304$ and $g(8) = 682$; whether $\mathrm{ex}(Q_7, C_4) = g(7)$ and $\mathrm{ex}(Q_8, C_4) = g(8)$ remains exactly as open as Open Problems 1 and 2 above, since $\mathrm{ex}(Q_n, C_4)$ itself is not known exactly for $n = 7, 8$. (The equality through $n = 6$ does not mean the odd-square construction alone for $n = 7$: only 389 of the 19 866 released 304-edge Q_7 solutions are odd-square. The Emléktábla statement of the problem [13] also records the observation that in the small dimensions checked there, some extremal C_4-free example does satisfy the odd-square condition; the Q_7 data here are consistent with that and extend it to $n = 7$ in the only sense currently checkable – among the known, not proven extremal, 304-edge solutions, odd-square examples exist (389 of them), even though they are a small minority.) Does $g(n) = \mathrm{ex}(Q_n, C_4)$ also hold at $n = 7, 8$, and continue for $n = 9, 10, \ldots$ once $\mathrm{ex}(Q_n, C_4)$ is itself determined, or does the gap $\mathrm{ex}(Q_n, C_4) - g(n)$ become positive at some n? Laplante et al. [14] give general lower and upper bounds on the frustration index $|E(Q_n)| - g(n)$ that match only when n is a power of 4; whether $g(n) = \mathrm{ex}(Q_n, C_4)$ persists beyond $n = 8$ is, in their language, a question about when this frustration-index gap and the C_4-free extremal gap coincide.

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Use of generative AI. The author used Claude (Anthropic) for: (i) language editing of the abstract; (ii) assistance in writing the supplementary verification script (`verify.py`) used to confirm C_4-freeness of the released edge-list certificates; and, for this revision, (iii) retrieval and cross-checking of the primary sources cited in Section 6 and (iv) drafting assistance for the revised text. All definitions, constructions, theorems, computations, and analyses are the author's own, including the 256-bit spin configuration used in the Q_8 witness of Section 6.3, whose provenance is recorded there as a dated correspondence account. The author reviewed all AI-assisted output, independently executed the verification and witness scripts against the released data, and takes full responsibility for the accuracy and integrity of the work. Because `verify.py` is both AI-assisted and the release's primary certificate checker, the certificate checks are additionally covered by a second implementation, `cross_verify.py`, now bundled with the release: it shares no code and no core mechanism with `verify.py` (bitmask common-neighbour counting over Hamming-distance-2 pairs instead of square enumeration by corner tuples, and a sorted-tuple set instead of hashing for the catalogue-level distinctness of the 19 866 edge sets), and it reproduces the same verdicts on every released certificate. That re-implementation is itself AI-assisted, so it mitigates the single-implementation risk rather than adding a human-written check. (An earlier ad-hoc re-implementation, reported at this point by the previous revision, was not preserved in the release and was therefore unauditable, as a round of review noted; the bundled script supersedes it.) The specific model version(s) of Claude used across the original submission and this revision were not recorded at the time and cannot be confirmed retrospectively; the disclosure above is accurate as to which tasks AI assistance was used for, but not as to the exact model version for each. The DPTV79 and MPR95 source retrieval and passage-level cross-checking in this revision were AI-assisted. During the final revision the cited primary-source PDF passages were checked directly at the page level: DPTV79 pp. 620–622 for the $h \geq 0$ restriction and the first lower bound, Table I, the H_6 construction, field multiplicities, and overfrustrated-plaquette count, and MPR95 pp. 3–4 of the arXiv preprint (cond-mat/9410089; the journal pagination is *J. Phys. A* **28**, 327–334) for the $D = 8, 9, 10$ energy statements and Hamiltonian normalisation. In this revision the LNSW26 preprint was additionally obtained and read in full (AI-assisted): its Theorem 5.2 statement and attribution and its Theorem 5.4 bound were checked verbatim against the Introduction and Section 6, its "at most" wording for the lower bound was confirmed on the page, and the $n \leq 6$ value list was confirmed *absent* from the preprint (see Section 6); these preprint checks were carried out on 25 August 2026 against HAL `hal-04080411`. On 26 August 2026 we additionally obtained and read the published version's full text (ACM Trans. Comput. Theory **18**(1), Art. 6, pp. 1–25, doi:10.1145/3777401; AI-assisted retrieval): its Section 5 numbering (Theorem 5.2, Theorem 5.4) matches the preprint's, its Theorem 5.4 proof states the negative-4-cycle property of the BHN construction as quoted in Section 6.2, and the $n \leq 6$ value list is absent from the published version as well. This disclosure should not be read as a claim that the author independently re-transcribed every quoted passage without AI assistance.

Disclosure statement

No potential conflict of interest was reported by the author. One related involvement is disclosed for completeness: the author performed an independent certificate-level check, by sparse cherry counting, of the Q_9–Q_{15} C_4-free edge lists contributed by R. Wrona to the

Erdős Problems discussion thread for problem 86 [12], confirming SHA-256 agreement, edge counts, validity of the hypercube edges, absence of loops and duplicates, and absence of C_4s. That check makes no claim of novelty or optimality for those certificates, and none of them is used in any argument here; the involvement is recorded because Open Problems 7 and the minimum-degree question above refer to the $n \geq 9$ range those certificates occupy.

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